

FMCG ERP SYSTEM

FMCG ERP Manufacturing Manual

Production · Planning · Quality · Compliance · Shop Floor

Audience: Production Managers, Quality Officers, Planning Managers, Shop Floor Supervisors

Version: v2.0 — Post-Recovery Edition

Date: 24 May 2026

Status: Confidential — Internal Use Only

ERP button/link recovery complete. All navigation targets restored and verified. Smoke test: 141/141 routes passed. BVT: 0. Broken action cards: 0.

FMCG ERP — Manufacturing Module Manual v2

Version: 2.0 (post-recovery)

Date: 2026-05-24

Audience: Production Managers, Quality Officers, Planning Managers, Shop Floor Supervisors, NPD Teams

Modules Covered: Recipes · BOM & Formula · Production · Planning · Shop Floor · NPD · Quality · Compliance

Table of Contents

1. Module Overview
 2. Recipes
 3. Recipe Bulk Import
 4. BOM & Formula
 5. Production
 6. Advanced Planning & MRP
 7. Shop Floor Operations
 8. New Product Development (NPD)
 9. Quality Control
 10. Compliance & Labelling
 11. Common Mistakes & Troubleshooting
 12. Related Modules
-

1. Module Overview

What it does: The Manufacturing module manages the full production lifecycle — from product formulation through to shop floor execution, quality control, and regulatory compliance.

Who uses it:

- Production Manager — creates production plans and work orders
- Quality Officer — logs inspections, manages certificates and complaints

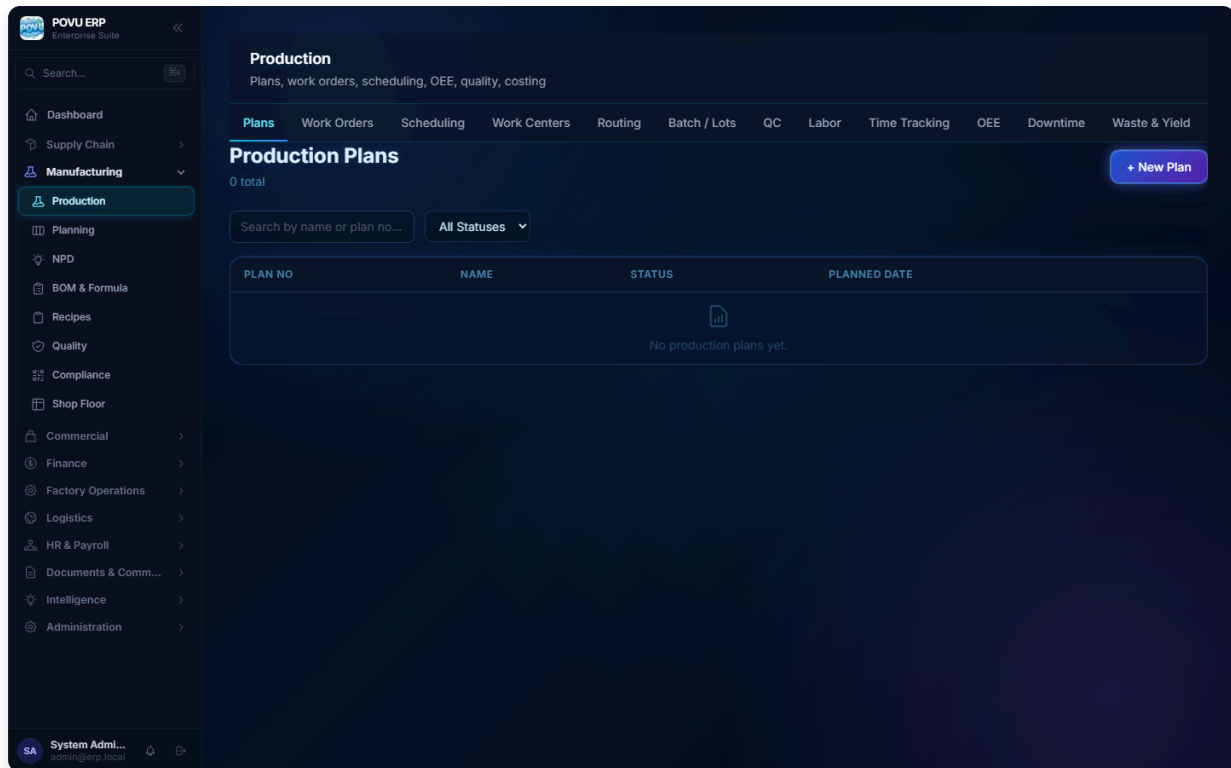
- Planning Manager — runs MRP/MPS and capacity planning
- Shop Floor Supervisor — monitors execution, records downtime and handovers
- NPD Manager — manages new product development stage-gates

When to use it:

- When setting up a new product formulation (Recipes → BOM)
- When planning a production run (Production Plans → Work Orders)
- When executing a production order on the factory floor (Shop Floor)
- When performing quality checks on finished goods or in-process batches
- When tracking OEE, downtime, and yield

Modules at a glance:

Module	Route	Purpose
Recipes	/dashboard/recipes	Formulation management
BOM	/dashboard/bom	Bill of materials and formula
Production	/dashboard/production	Plans, orders, scheduling, OEE
Planning	/dashboard/planning	MRP, MPS, capacity scheduling
Shop Floor	/dashboard/shop-floor	Real-time terminal and supervisor
NPD	/dashboard/npd	New product stage-gate
Quality	/dashboard/quality	QC, certificates, complaints
Compliance	/dashboard/compliance	GS1 labels, regulatory certs



Production module main view.

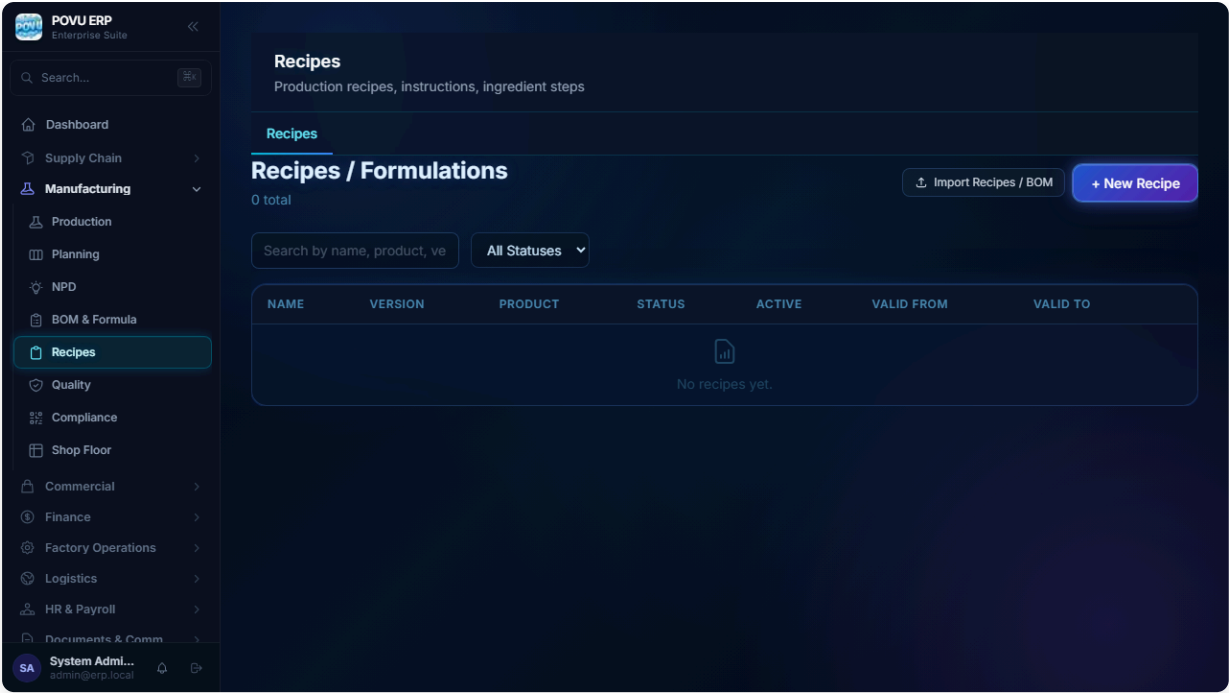
2. Recipes

Route: `/dashboard/recipes`

Required permission: `recipe.view`

What it does

Manages product formulations — the ingredients (BOM items) and process parameters (temperature, pH, mixing time) for each product version. A recipe must be **Approved** before it can be used in production.



Recipes list page with search, status filter, Import, and New Recipe buttons.

Tabs and Sections

The Recipes page is a single list (no workspace tabs). Each recipe row links to a detail page.

Every Button Explained

Button	Where	What it does
+ New Recipe	Top right	Opens New Recipe modal
Import Recipes / BOM	Top right	Opens 3-tab CSV import modal
Delete (trash icon)	Row action	Deletes DRAFT recipe only

Search and Filters

- **Search bar** — filters live across: recipe name, product SKU, product name, version number
- **Status filter** — filter by DRAFT / APPROVED / OBSOLETE

Status Values

Status	Meaning
DRAFT	Under development; editable and deletable
APPROVED	Approved for production; no edits allowed
OBSOLETE	Retired; no longer active

Status flow: **DRAFT** → **APPROVED** → **OBSOLETE** (no reverse)

Creating a Recipe

1. Click **+ New Recipe**
2. Fill in the modal:

Field	Required	Notes
Product	Yes	Select from products master
Recipe Name	Yes	Free text
Version	Yes	Default "1.0"
Description	No	Not shown in list
Valid From / Valid To	No	Validity window

3. Click **Create** — recipe created as DRAFT

New Recipe modal. Product, Name, and Version are required.

Recipe Detail Page

Route: `/dashboard/recipes/{recipe_id}`

Two sub-sections on detail page:

BOM Items (Ingredients)

Field	Description
Material	Link to materials master
Line No	Ordering sequence (must be unique)
Quantity	Amount per batch
Unit	KG, L, G, ML, etc.
Loss %	Expected process loss (default 0)
Optional	Marks non-mandatory ingredients
Alt Group	Groups substitutable items
Notes	Free text per line

Process Parameters (Steps)

Field	Description
Step No	Sequence (must be unique)
Step Name	e.g. Blending, Pasteurisation
Temp (°C)	Target process temperature
pH	Target pH value
Viscosity (cP)	Target viscosity
Mix Time (min)	Mixing duration
RPM	Agitator speed
Notes	Free text

DRAFT-only rule: BOM items and process parameters can only be added/edited/deleted while status is DRAFT.

Required Data Before Starting

- Products must exist in the Products master
- Materials must exist in the Materials master
- At minimum one BOM item needed before recipe can be approved

What Happens After Save

- Recipe created as DRAFT — visible in list
- Detail page opens automatically
- Add BOM items and process parameters before approving

Common Mistakes

- Trying to edit an APPROVED recipe → blocked by API (HTTP 422)
 - Forgetting to set version (defaults to "1.0" — acceptable but not distinguishable from other v1 recipes)
 - Deleting an APPROVED recipe → delete button not shown for APPROVED/OBSOLETE
-

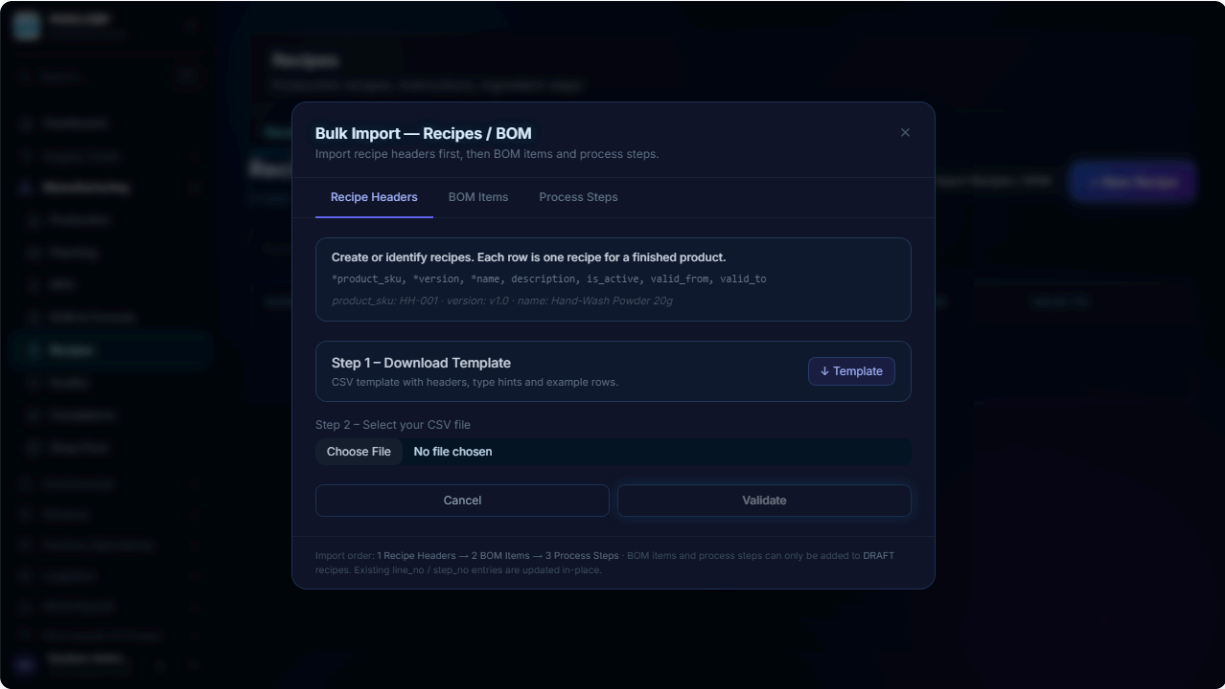
3. Recipe Bulk Import

Button: Import Recipes / BOM on the Recipes page

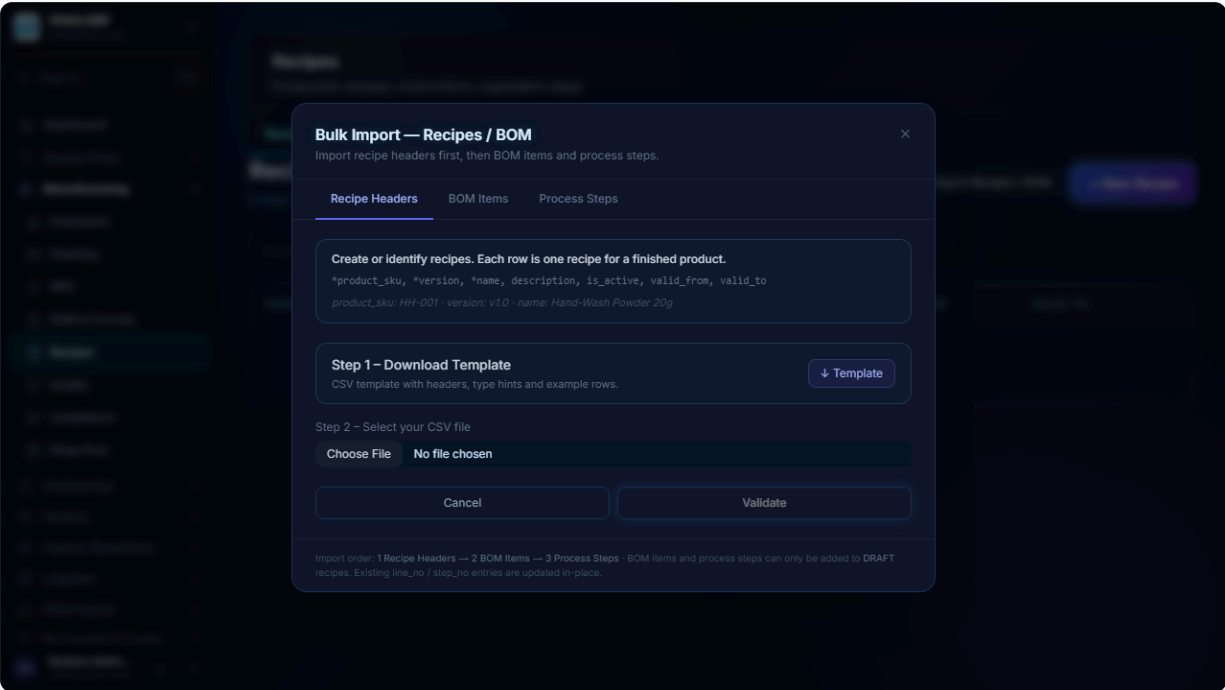
When to Use

When migrating recipes from Excel or another system. Faster than creating one by one.

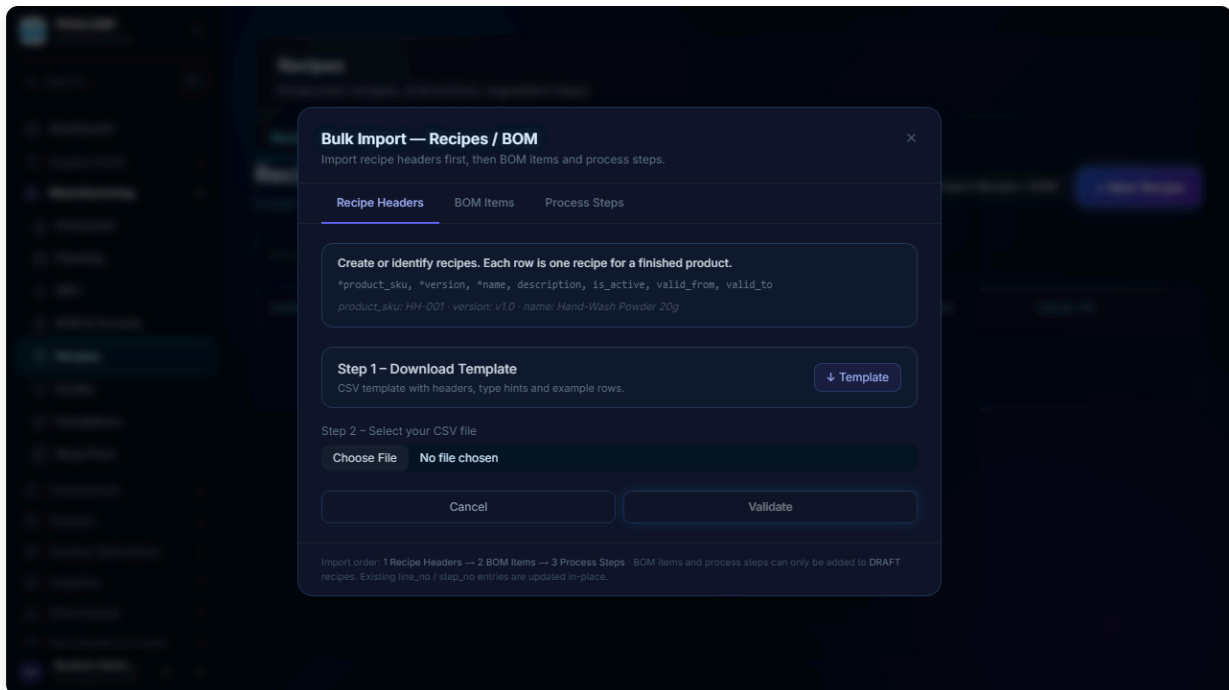
Import Modal — 3 Tabs



Tab 1: Recipe Headers. Upload the CSV with one row per recipe.



Tab 2: BOM Items. Upload BOM lines keyed by recipe name or ID.



Tab 3: Process Steps. Upload process parameter rows.

Import Steps (in order)

1. **Tab 1 — Recipe Headers:** Upload CSV with columns: `name, product_sku, version, description, valid_from, valid_to`
2. **Tab 2 — BOM Items:** Upload CSV with columns: `recipe_name, material_code, line_no, quantity, unit, loss_percentage, is_optional, alternative_group, notes`
3. **Tab 3 — Process Steps:** Upload CSV with columns: `recipe_name, step_no, step_name, target_temperature, target_ph, target_viscosity, mixing_time_minutes, rpm, notes`

Order matters: Import headers first. BOM and process steps reference recipe names created in Tab 1.

Validation Rules

- All referenced products (by SKU) must exist in the system
- All referenced materials (by code) must exist in the system
- Duplicate recipe name + version combinations are rejected
- Numeric fields (quantity, temperature, pH) must be valid numbers

What Happens After Import

- New recipes created as DRAFT
- Validation errors shown inline per row
- Successful rows imported; failed rows shown with error message

4. BOM & Formula

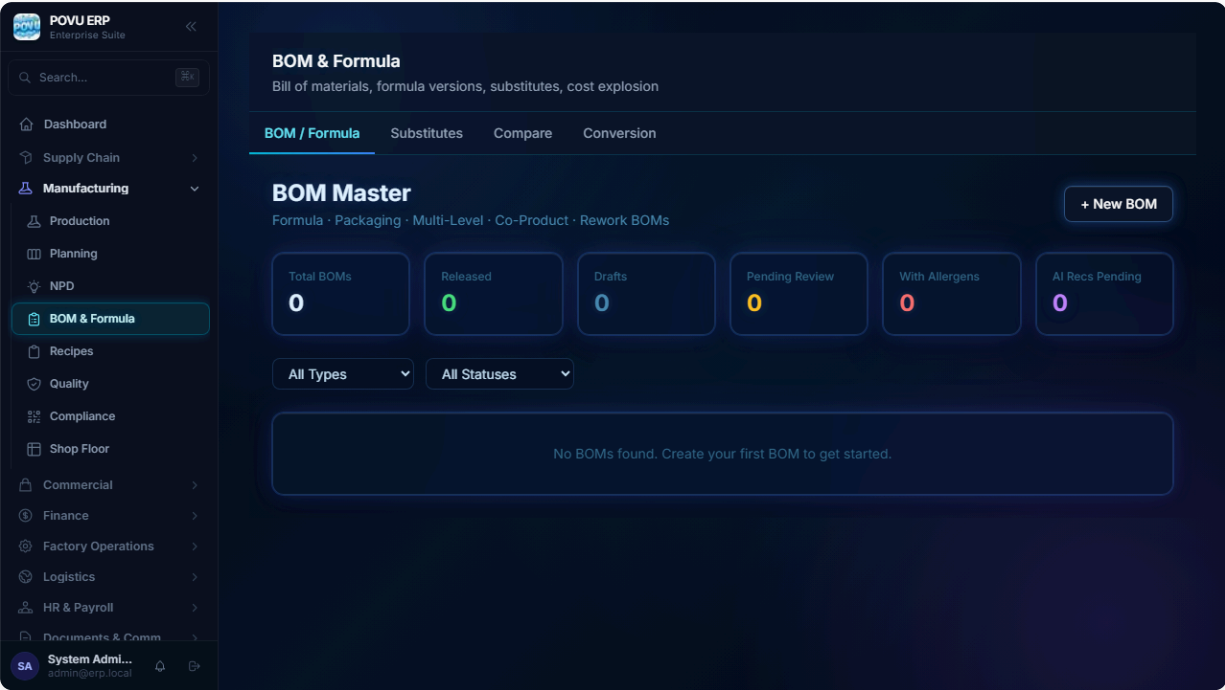
Route: `/dashboard/bom`

Required permission: `bom.view`

What it does

Manages Bills of Materials — the formal list of components required to produce a finished good. BOM is the production-facing view; Recipe is the formulation-facing view.

Tabs

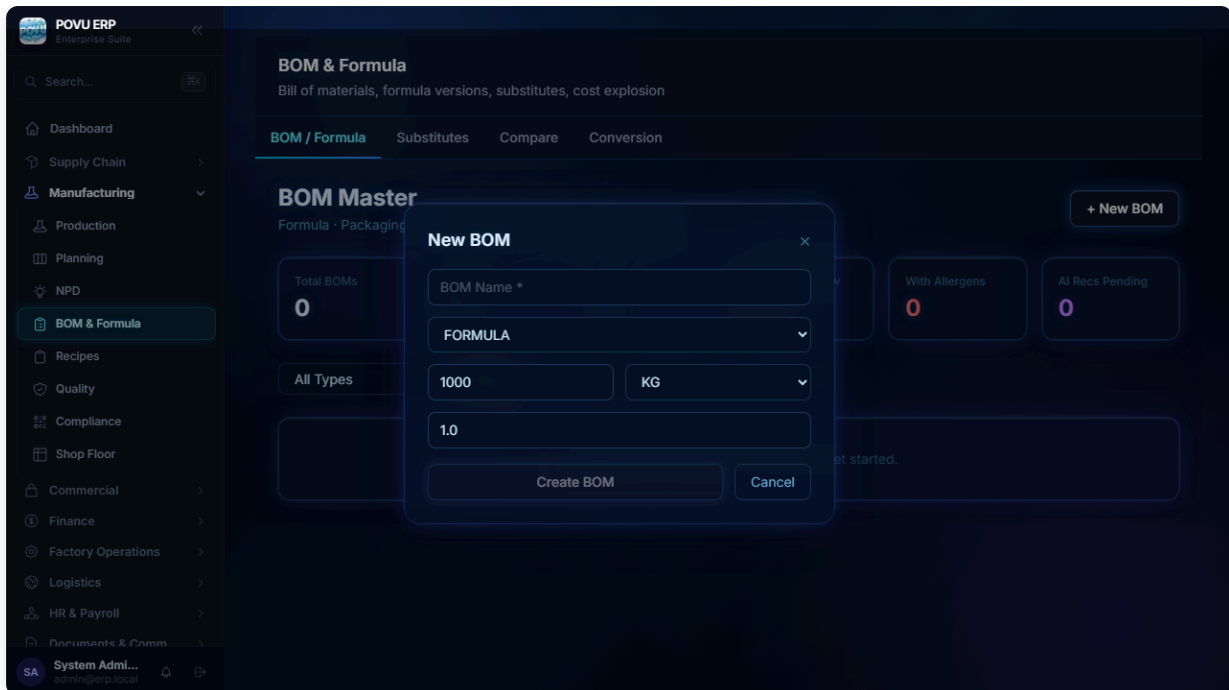


BOM List tab showing all BOMs with product, version, and status columns.

Tab	Purpose
BOM List	All BOMs, searchable and filterable
Substitutes	Substitute groups and alternative components
Compare	Side-by-side BOM version comparison
Conversion	Unit conversion factors

Creating a BOM

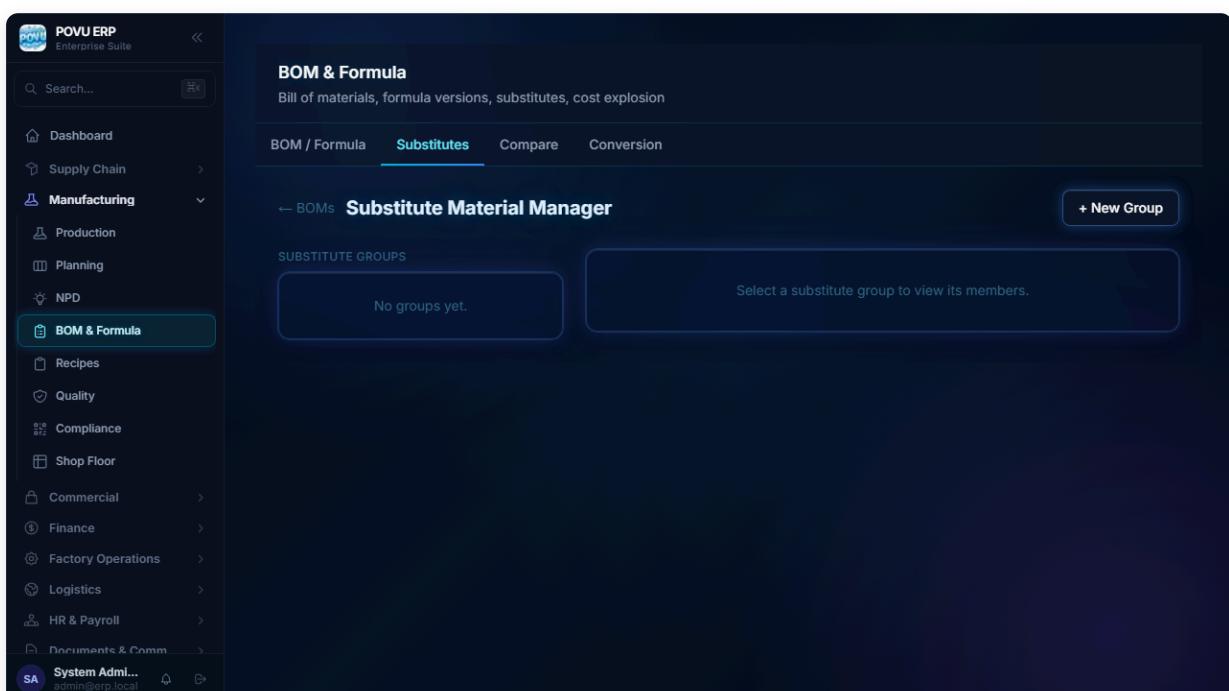
Click **+ New BOM** (requires `bom.create`).



New BOM modal with product, version, and effective date fields.

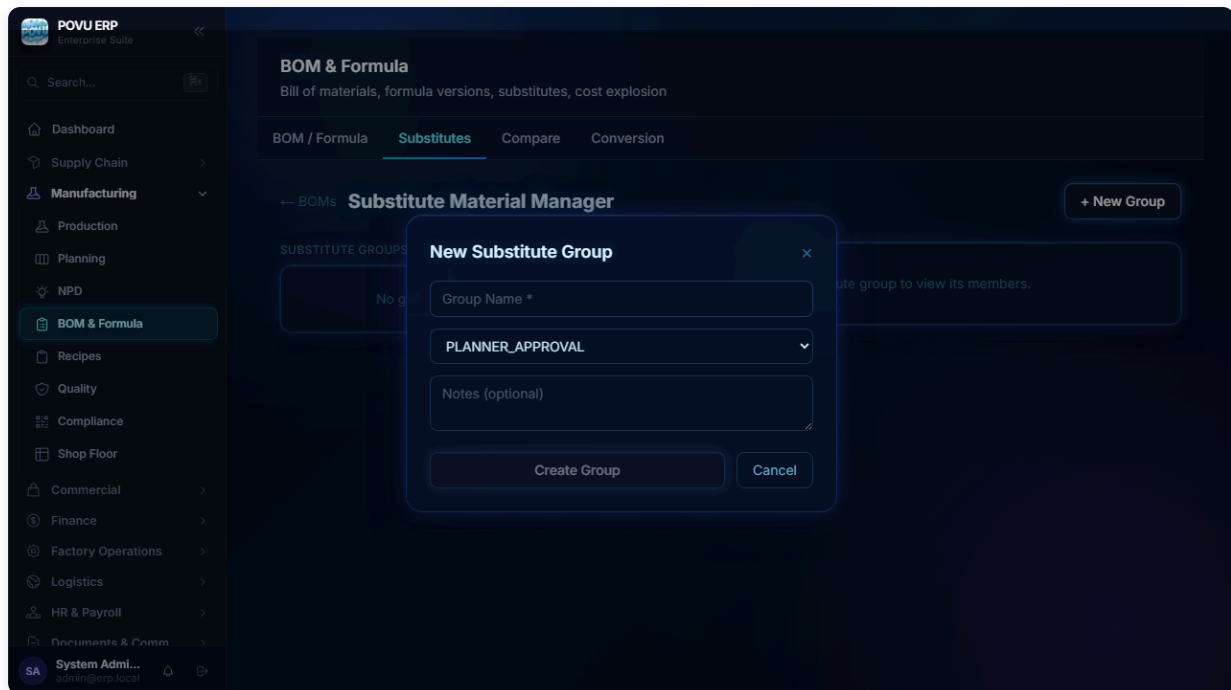
Field	Required	Notes
Product	Yes	Select from products
BOM Type	Yes	Manufacturing / Kit / Service
Version	Yes	e.g. "1.0"
Effective Date	No	Date from which BOM is active

BOM Substitutes Tab

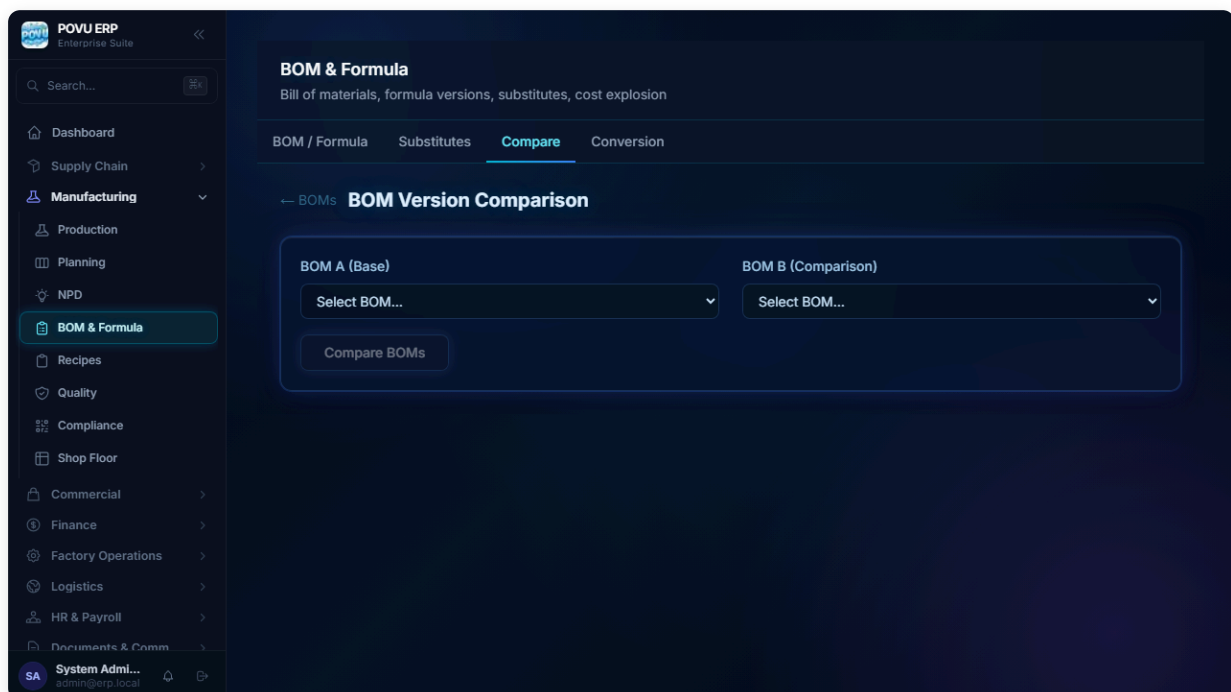


Substitute Groups — define alternative materials that can replace a primary component.

Click **+ New Substitute Group** to create a group:



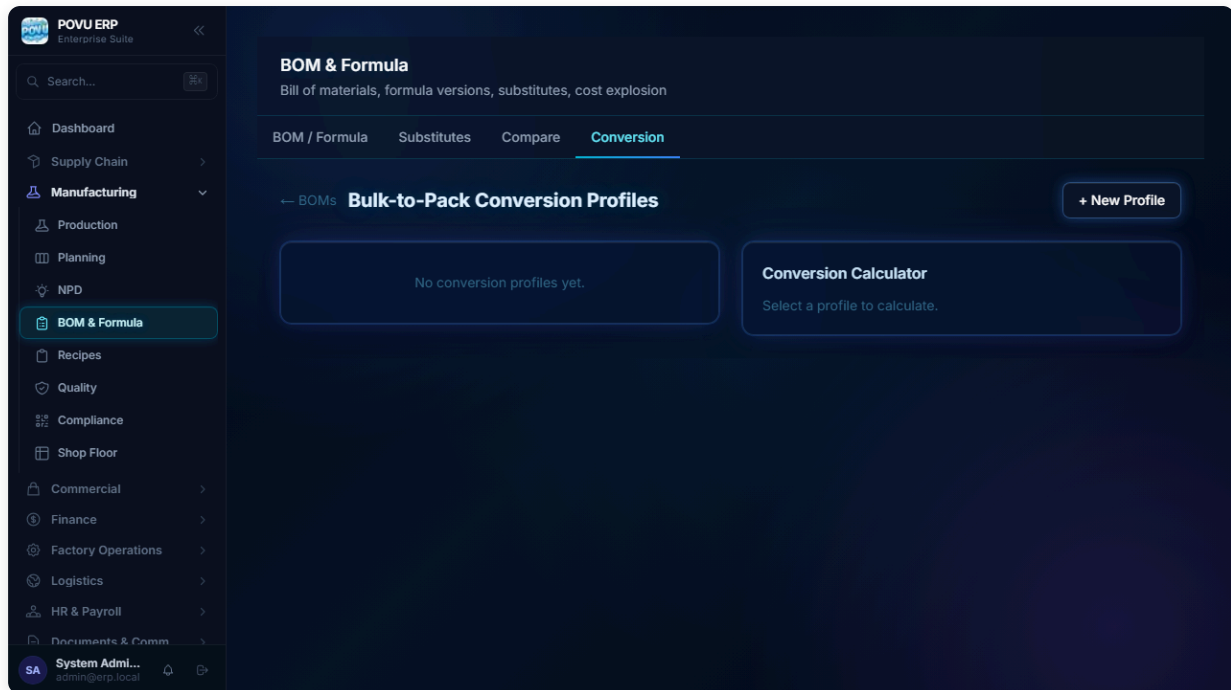
BOM Compare Tab



Side-by-side comparison of two BOM versions. Highlights additions, removals, and quantity changes.

Select two BOMs from the dropdowns and click **Compare**.

BOM Conversion Tab



Unit conversion factors used when BOM quantities differ from inventory units.

Common Mistakes

- Creating BOM before the product exists in the system
- Leaving version blank (causes duplicate-detection failures)
- Not setting Effective Date when multiple BOM versions exist for same product

5. Production

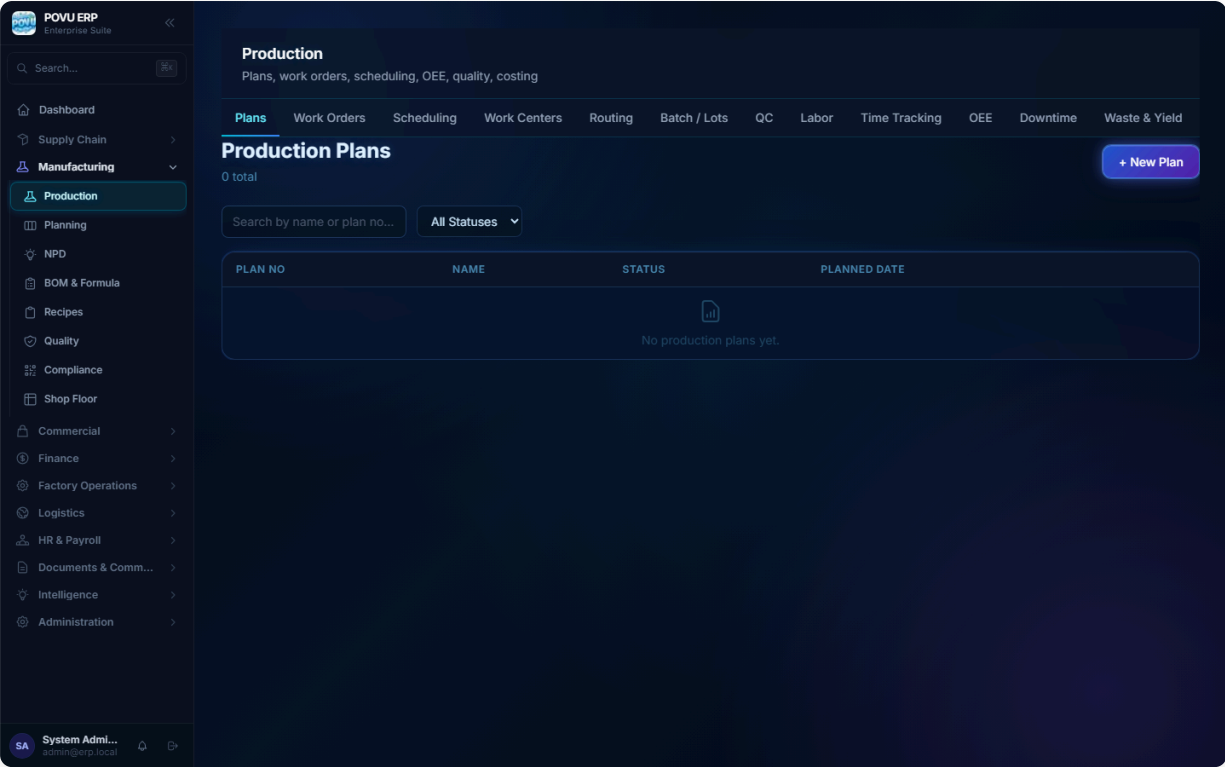
Route: `/dashboard/production`

Required permission: `production.view`

What it does

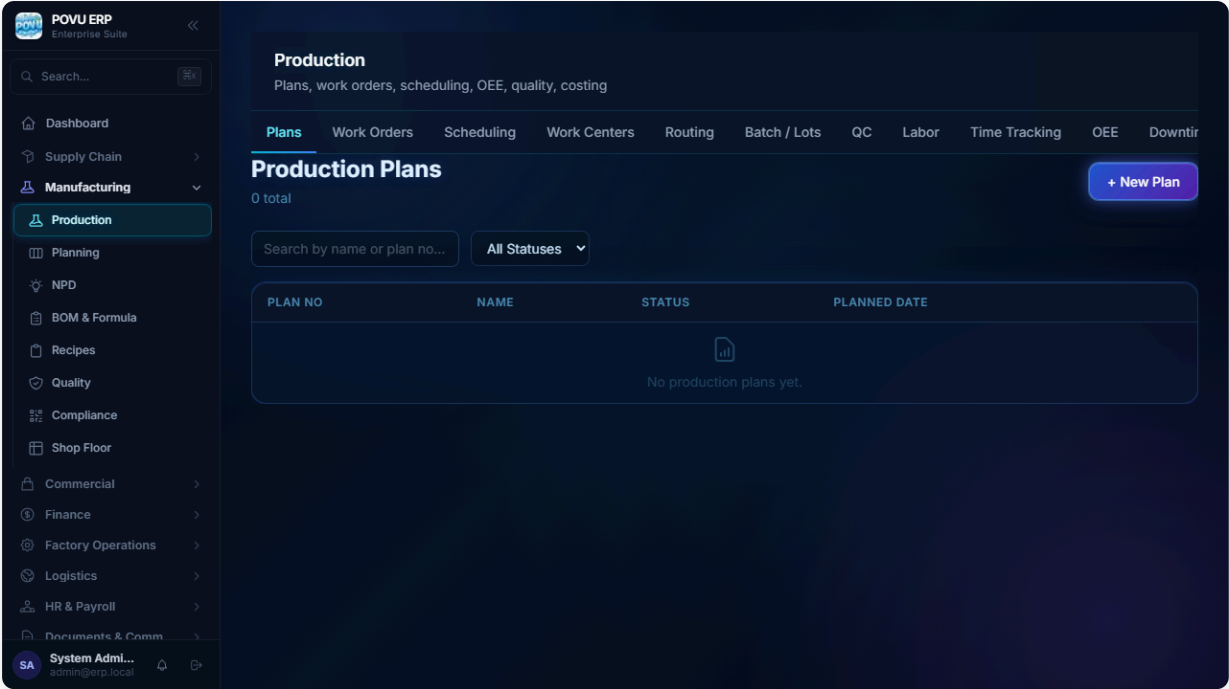
Central hub for production execution: create plans, raise work orders, schedule operations, track OEE, record costing and variance.

Tabs (20 total)



Tab	Purpose
Plans	Production plans (multi-order groupings)
Work Orders	Individual production orders
Scheduling	Gantt-style scheduling and sequencing
Work Centers	Machine and labor centers configuration
Routing	Operation sequences per product
Batch/Lots	Batch tracking and lot numbers
QC	In-process quality checks
Labor	Labor entries and time allocation
Time Tracking	Clock-in/out and operation time
OEE	Overall Equipment Effectiveness dashboard
Downtime	Downtime event log
Waste/Yield	Scrap, waste, and yield recording
WIP	Work-in-process inventory
Costing	Actual vs. standard cost per order
Variance	Cost and yield variance reports
Reports	Production summary reports
Execution	Real-time execution status
Machine Ops	Machine operation parameters
Material Flow	Material consumption vs. plan
Projects	Production project groupings

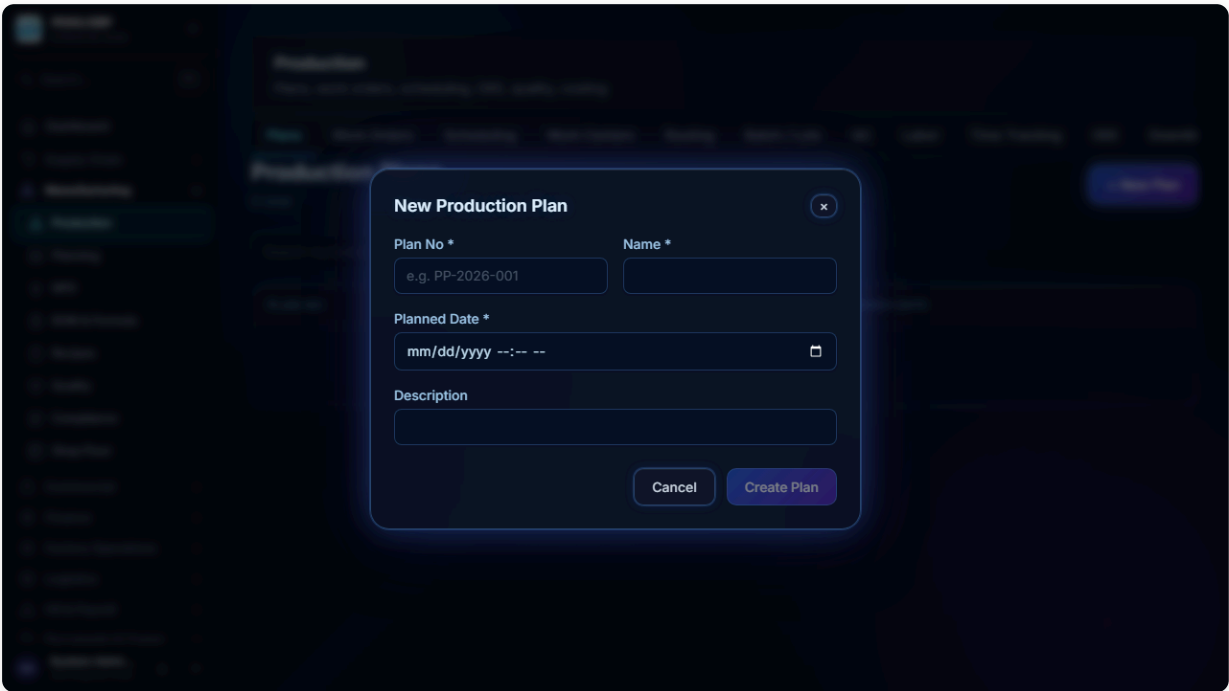
Production Plans Tab



Plans tab with plan list, status filter, and New Plan button.

Creating a Plan:

1. Click + New Plan
2. Fill modal:

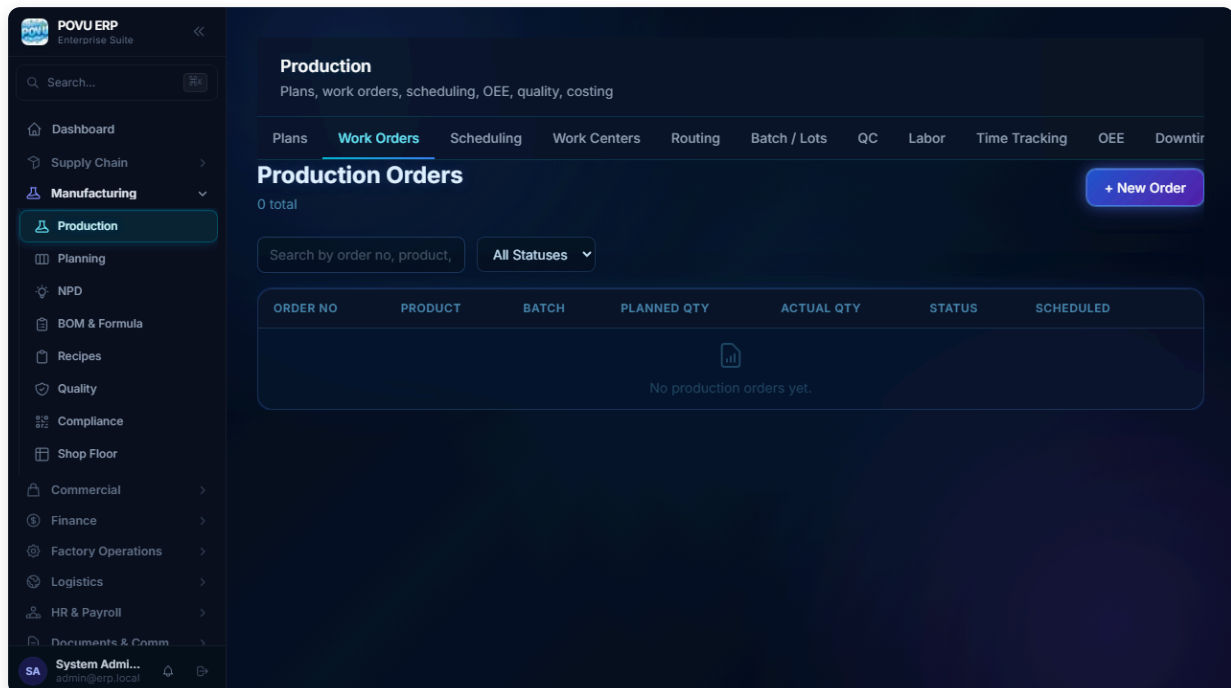


Field	Required	Notes
Plan Name	Yes	Free text
Start Date	Yes	Planned start
End Date	Yes	Planned completion
Product	No	Filter for product-specific plan
Notes	No	Free text

3. Plan created as DRAFT — add work orders from the Work Orders tab referencing this plan

Plan Status Flow: DRAFT → CONFIRMED → IN_PROGRESS → COMPLETED

Work Orders Tab



Work Orders tab showing all orders with status, product, quantity, and scheduled dates.

Creating a Work Order:

1. Click + New Work Order

New Production Order

Order No * Batch No

Product *

Recipe (Approved) *

Planned Quantity * Unit

FG Target Warehouse * Material Source Warehouse

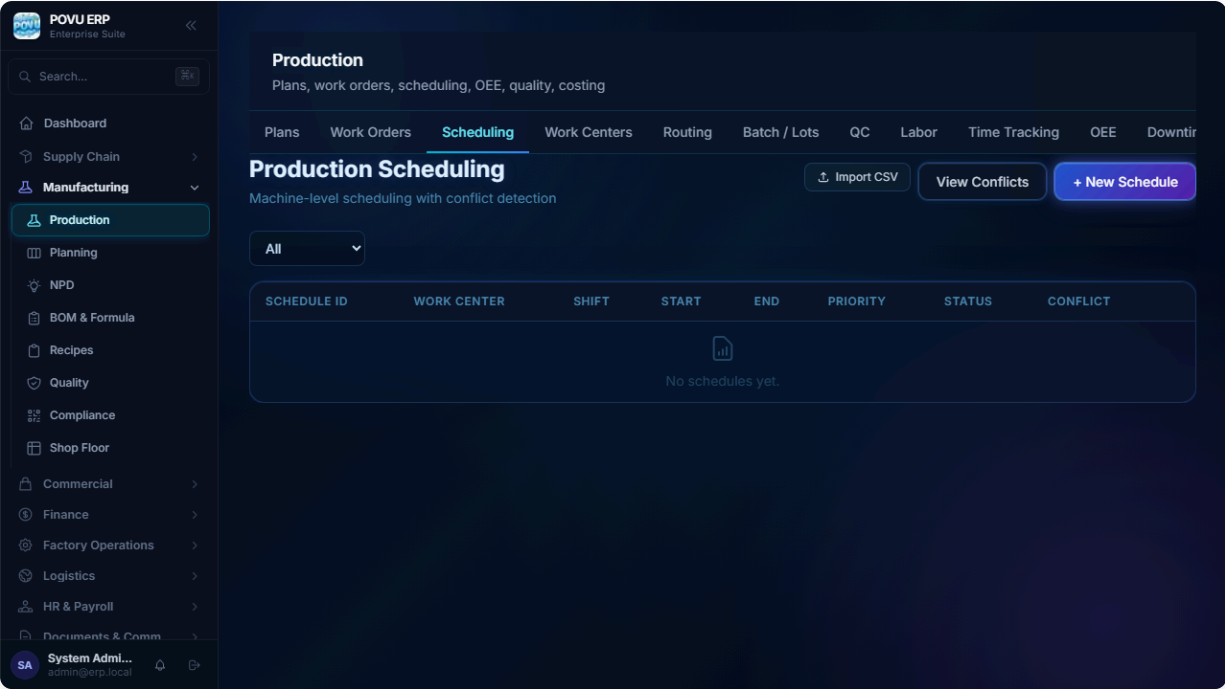
Scheduled Start * Scheduled End *

Notes

Field	Required	Notes
Product	Yes	Finished good to produce
Quantity	Yes	Planned quantity
Unit	Yes	UOM
BOM	Yes	Select approved BOM version
Recipe	No	Link to recipe for process params
Work Center	No	Primary work center
Planned Start	Yes	Scheduled start date/time
Planned End	Yes	Scheduled end date/time
Production Plan	No	Parent plan reference

Work Order Status Flow: DRAFT → CONFIRMED → RELEASED → IN_PROGRESS → COMPLETED → CLOSED

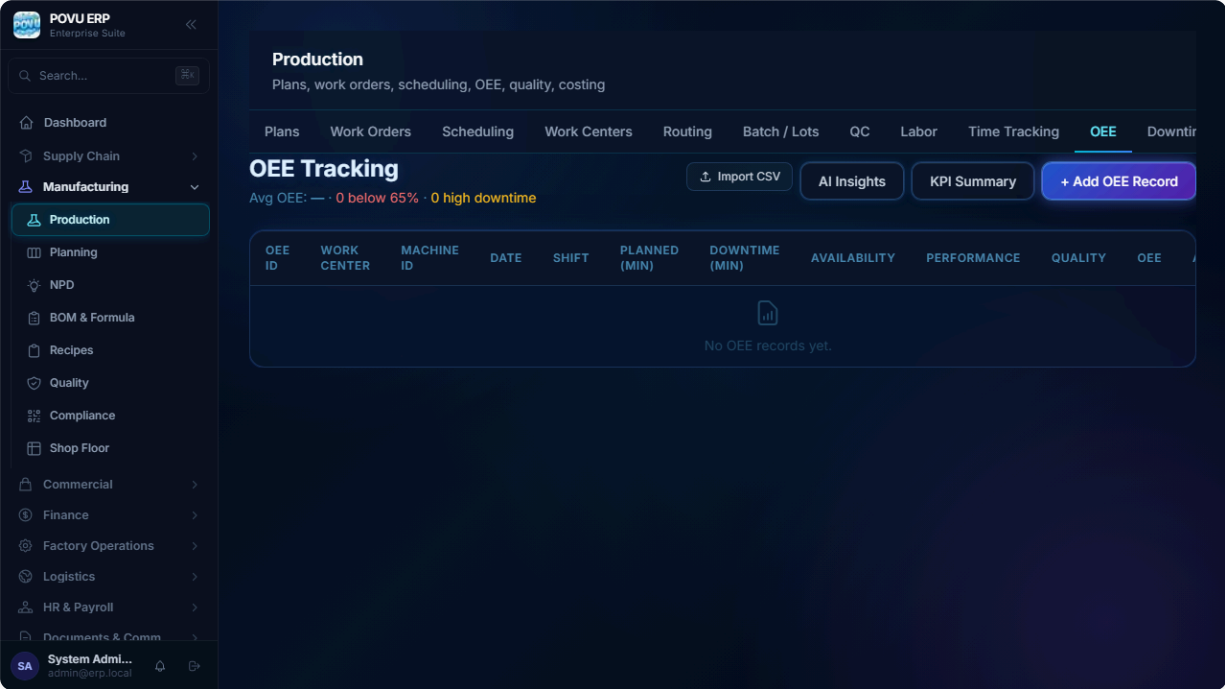
Scheduling Tab



Gantt chart view of all work orders by work center and date.

Drag-and-drop rescheduling. Click any bar to open work order detail.

OEE Tab



OEE dashboard showing Availability, Performance, Quality KPIs and trend charts.

OEE = Availability × Performance × Quality

KPI	Calculation
Availability	(Planned - Downtime) / Planned
Performance	Actual Output / Theoretical Max Output
Quality	Good Units / Total Units Produced

Downtime Tab

Production
Plans, work orders, scheduling, OEE, quality, costing

Plans Work Orders Scheduling Work Centers Routing Batch / Lots QC Labor Time Tracking OEE **Downtime**

Downtime Tracking [Import CSV](#) [+ Log Downtime](#)

Total downtime: 0 min (0 events)

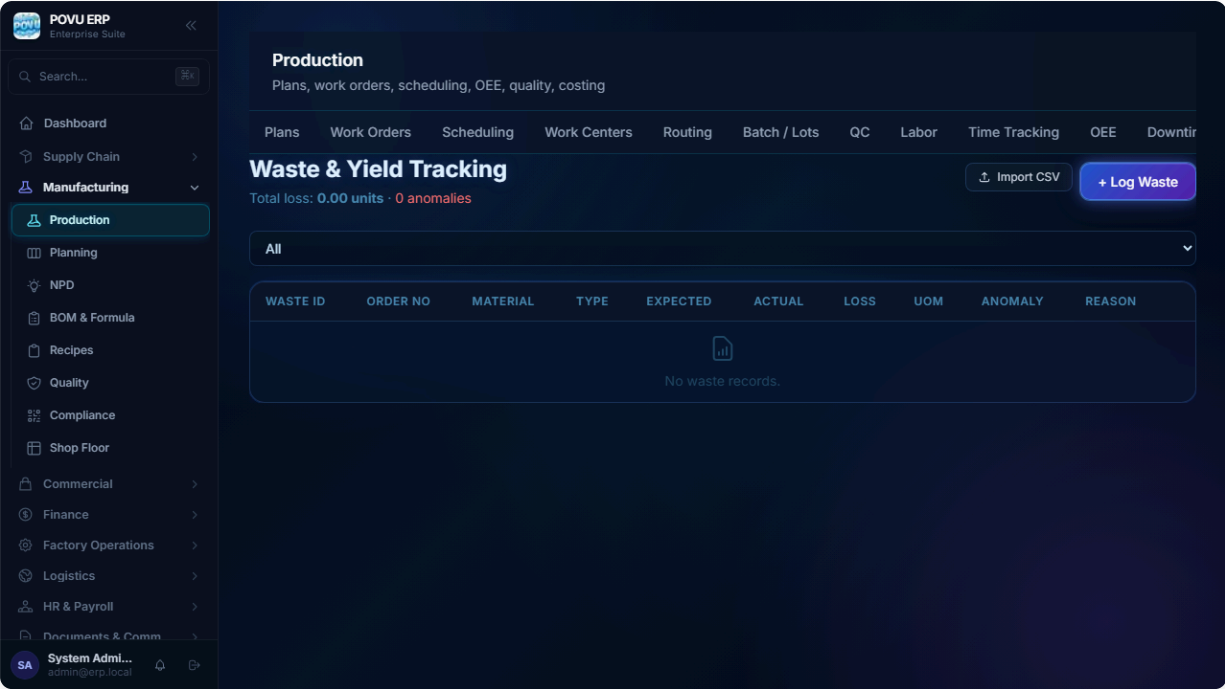
All

EVENT ID	WORK CENTER	MACHINE ID	START	END	DURATION (MIN)	CATEGORY	REASON	REPORTED BY
No downtime events.								

Downtime log with machine, category, duration, and reason columns.

Log downtime events: select machine, category (Mechanical, Electrical, Changeover, Break, etc.), start/end time, root cause notes.

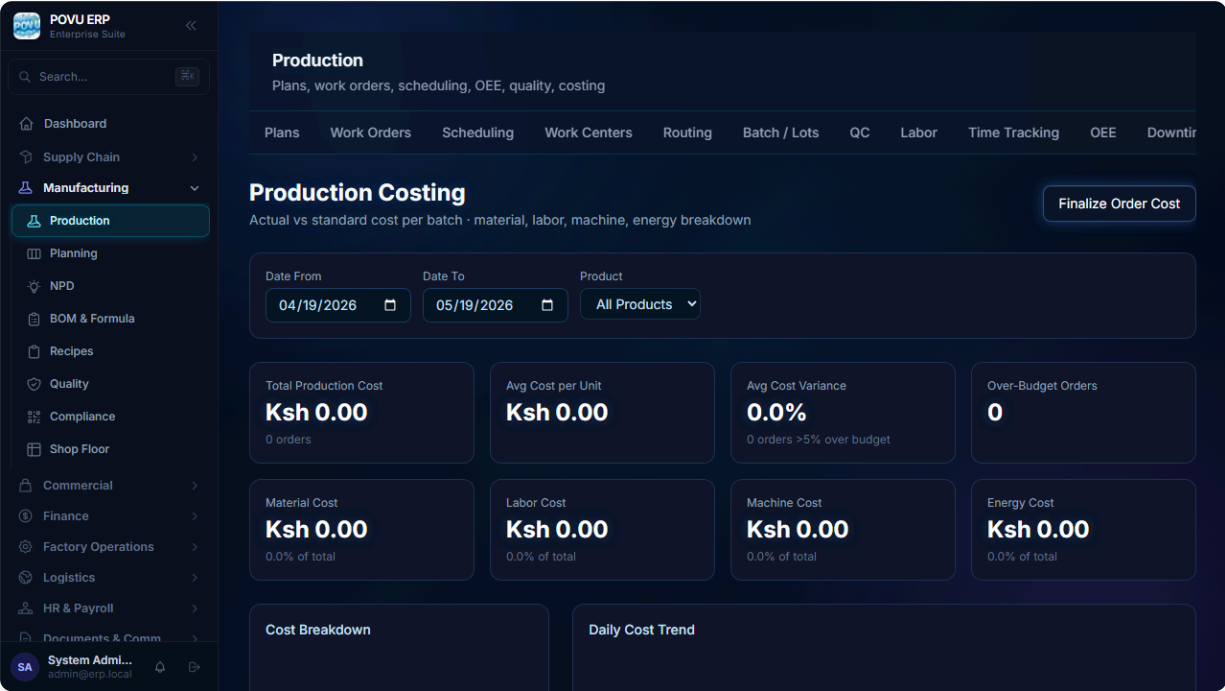
Waste/Yield Tab



Waste and yield recording per work order and material.

Record actual yield vs. planned yield. Waste entries create inventory adjustments automatically.

Costing Tab



Actual vs. standard cost breakdown by material, labor, and overhead.

Required Data Before Starting Production

- Products and materials in master data

- Approved BOM version for each product
- Work centers configured
- Routing defined (optional but needed for scheduling)

What Happens After Completing a Work Order

- Finished goods inventory increases by produced quantity
- Material consumption reduces raw material inventory
- Costing records are created (actual vs. standard)
- Batch/lot numbers assigned if batch tracking enabled

6. Advanced Planning & MRP

Route: `/dashboard/planning`

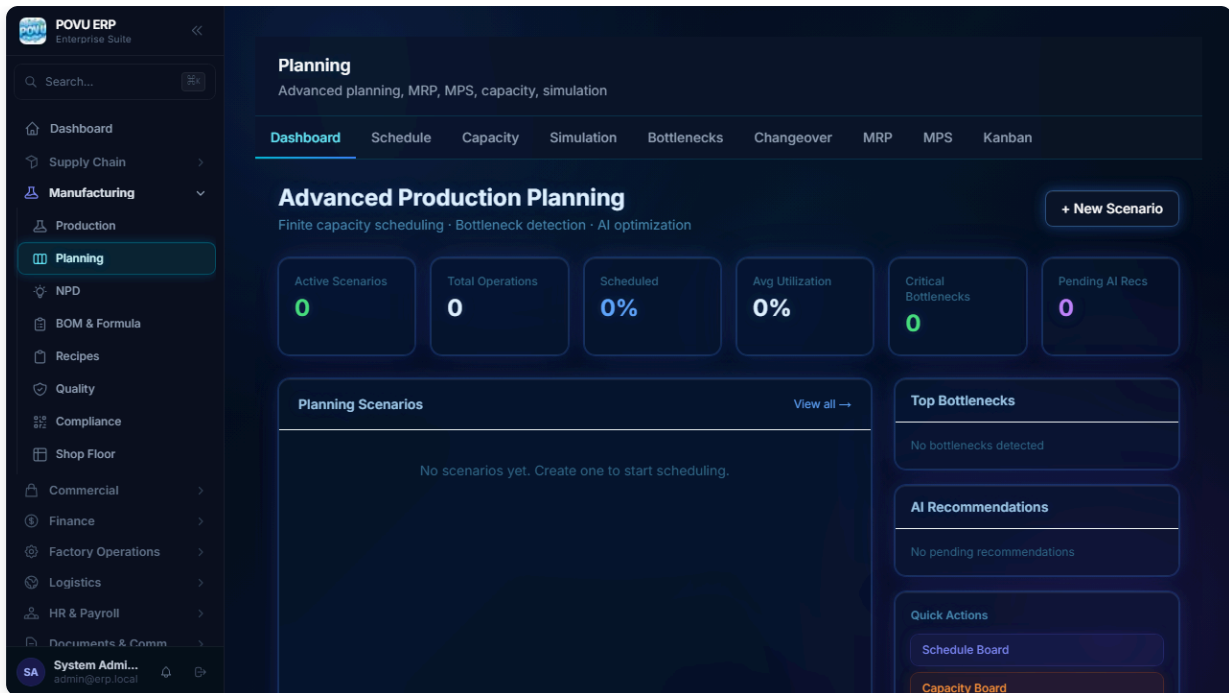
Required permission: `planning.view`

What it does

Finite capacity scheduling, Material Requirements Planning (MRP), Master Production Scheduling (MPS), and what-if scenario simulation.

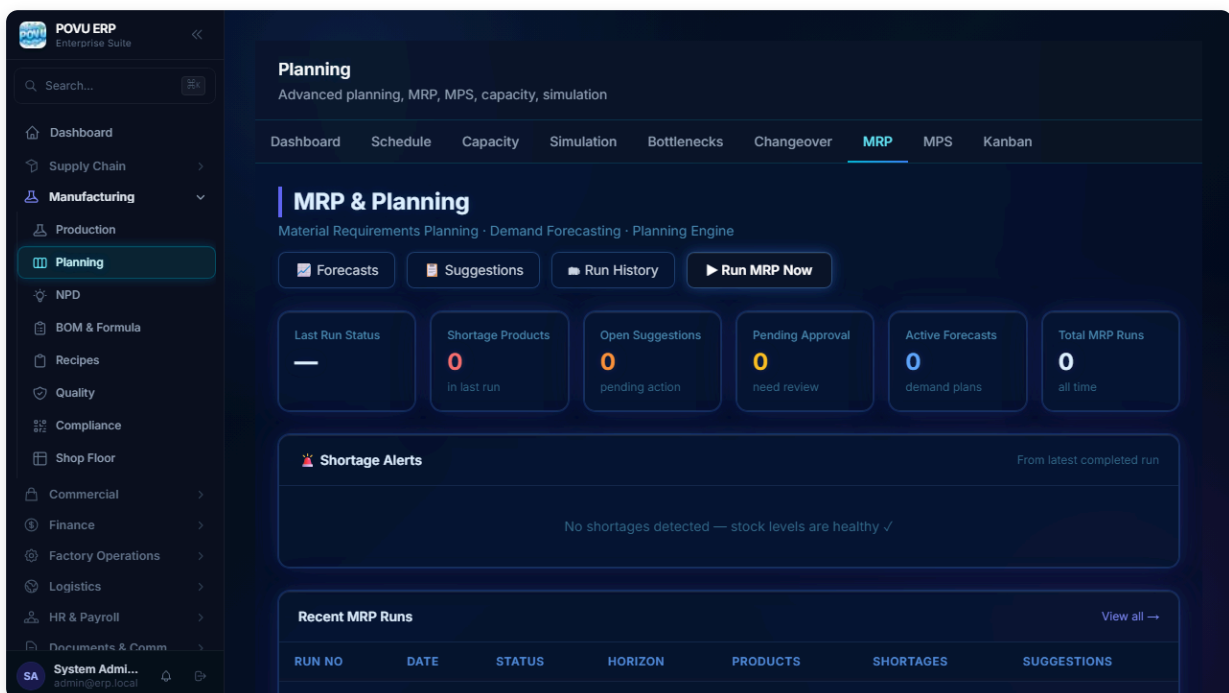
Tabs

Tab	Purpose
Dashboard	Planning KPI summary
Schedule	Finite capacity schedule
Capacity	Work center capacity profiles
Simulation	What-if scenario planning
Bottlenecks	Bottleneck identification
Changeover	Changeover time matrix
MRP	Material requirements planning run
MPS	Master production schedule
Kanban	Kanban card management



Planning module dashboard with KPI cards and schedule overview.

MRP Tab

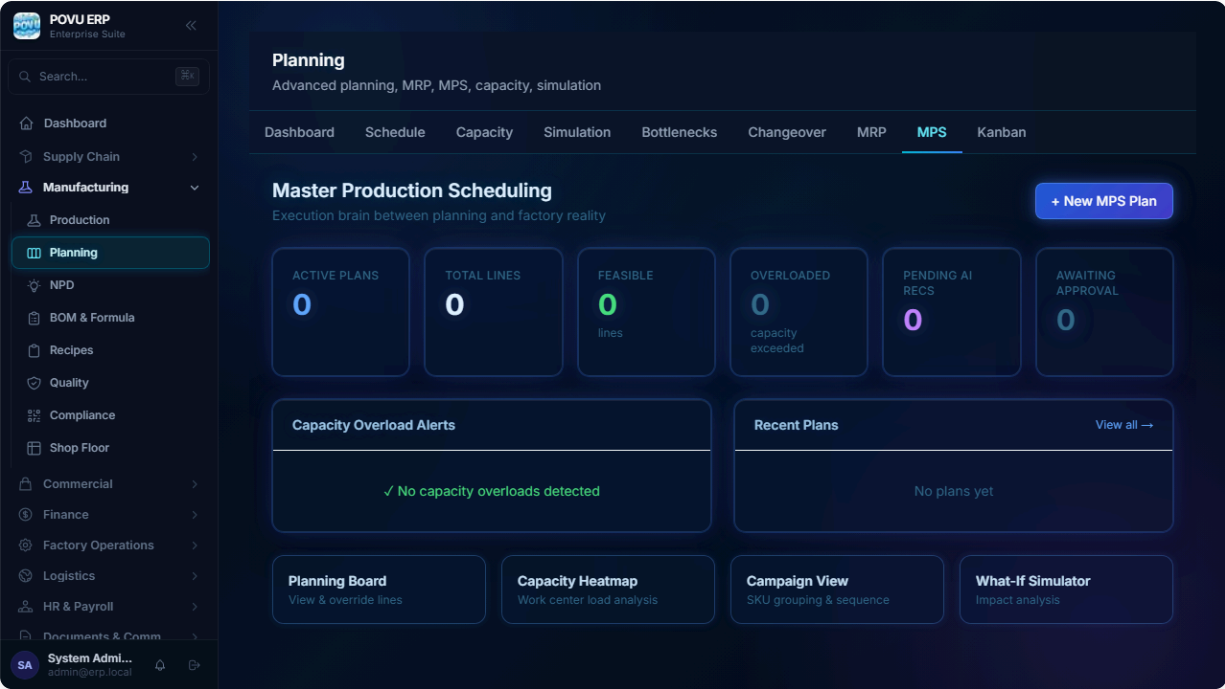


MRP run results — shows recommended purchase orders and production orders per material.

Running MRP:

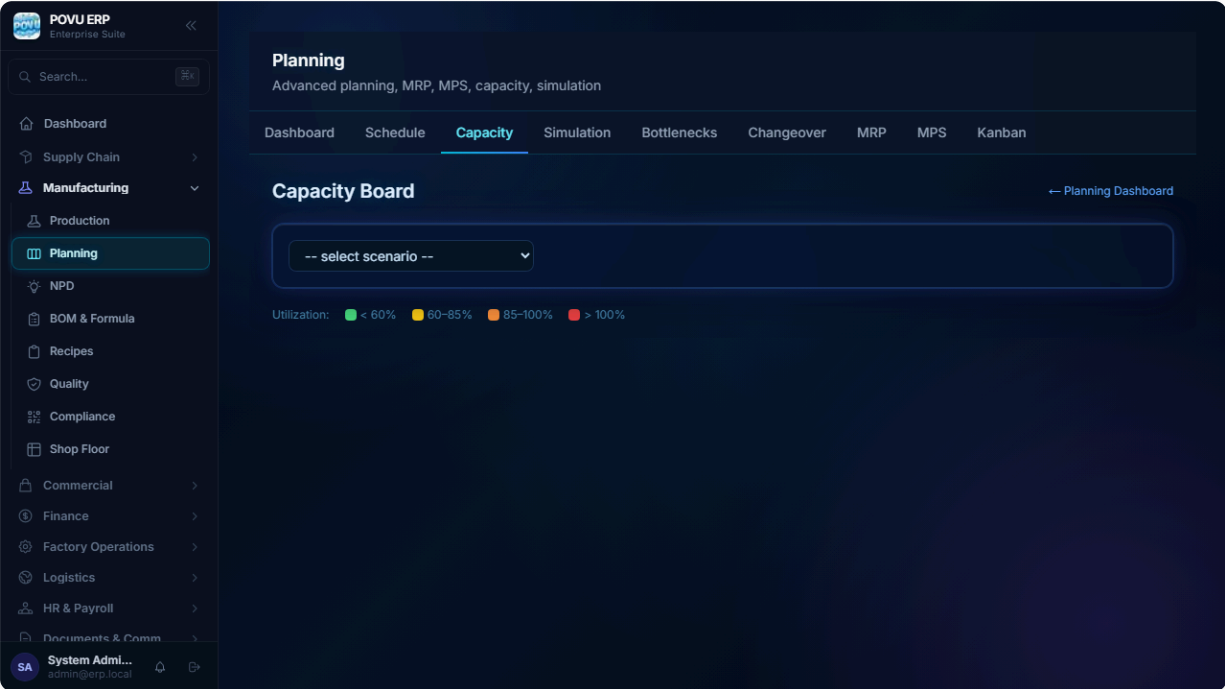
1. Go to Planning → MRP tab
2. Set horizon (default 30 days)
3. Click **Run MRP**
4. Review recommendations — approve to convert to purchase requisitions or work orders

MPS Tab



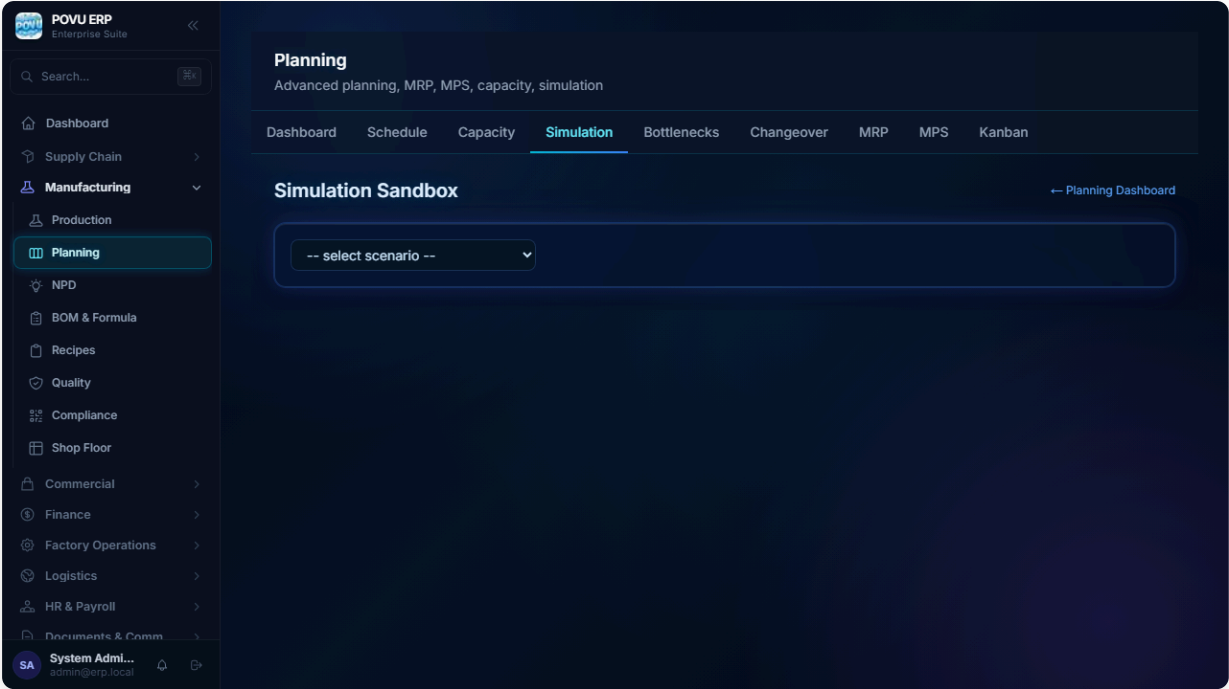
MPS showing demand vs. supply balance per period.

Capacity Tab



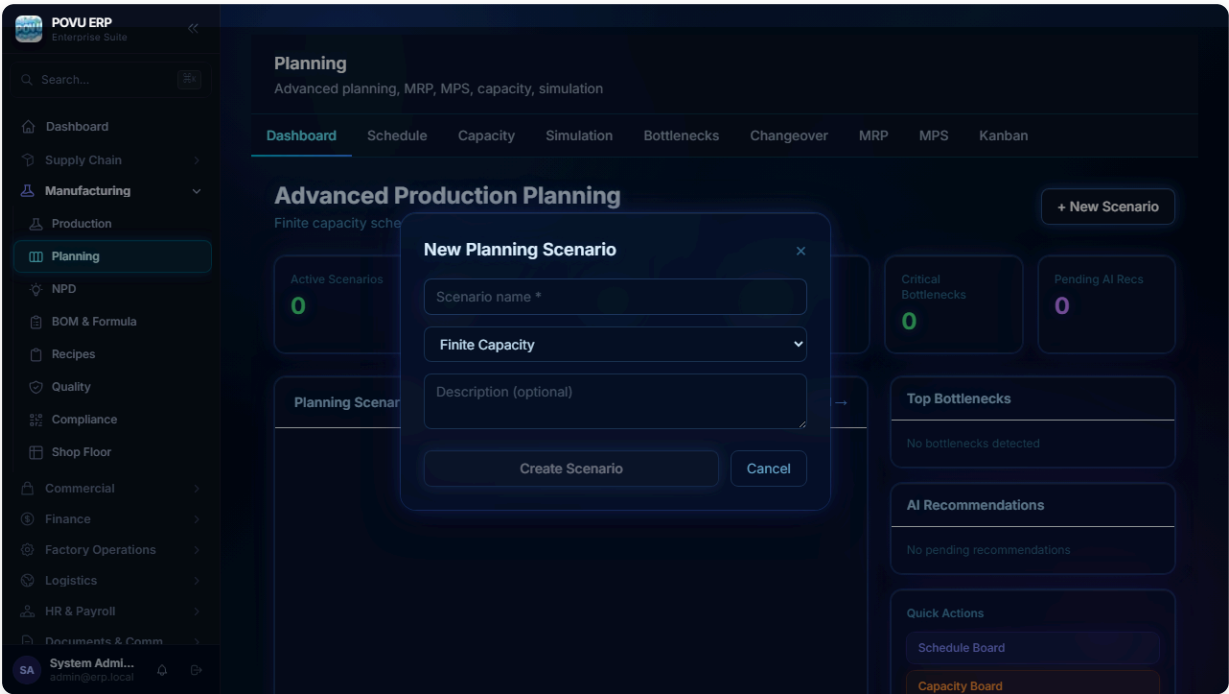
Work center capacity — planned load vs. available capacity by period.

Simulation Tab



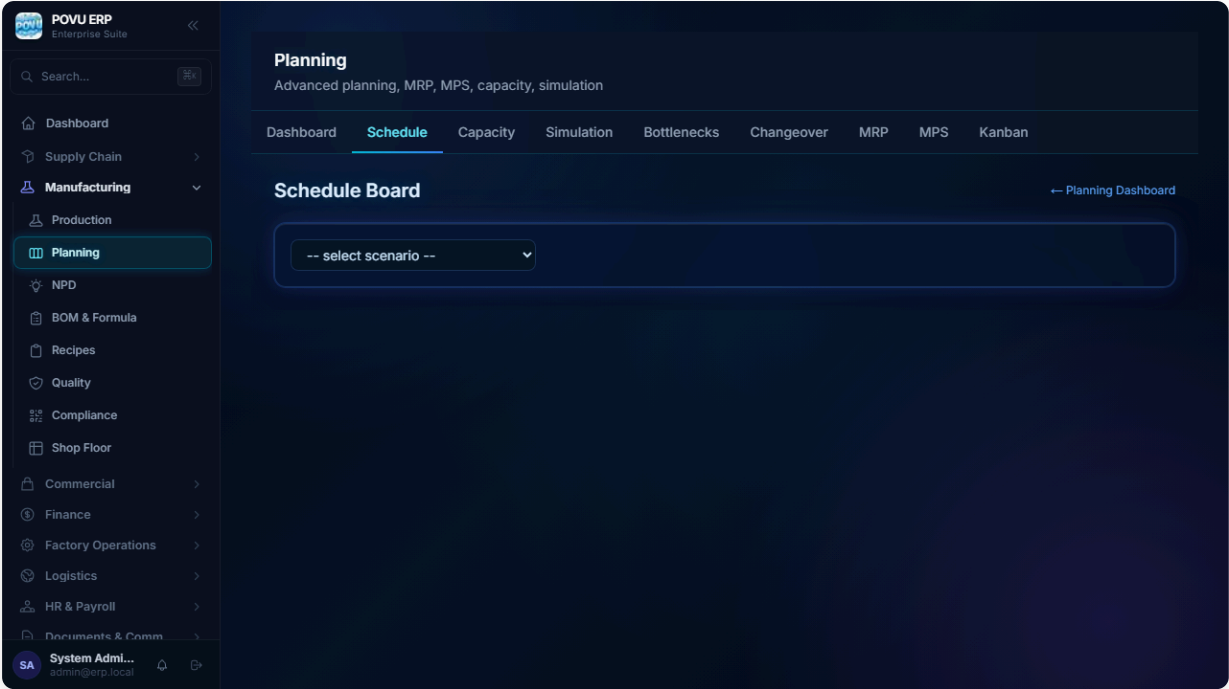
Create what-if scenarios to test schedule changes without affecting live data.

Click **+ New Scenario** to create a simulation:



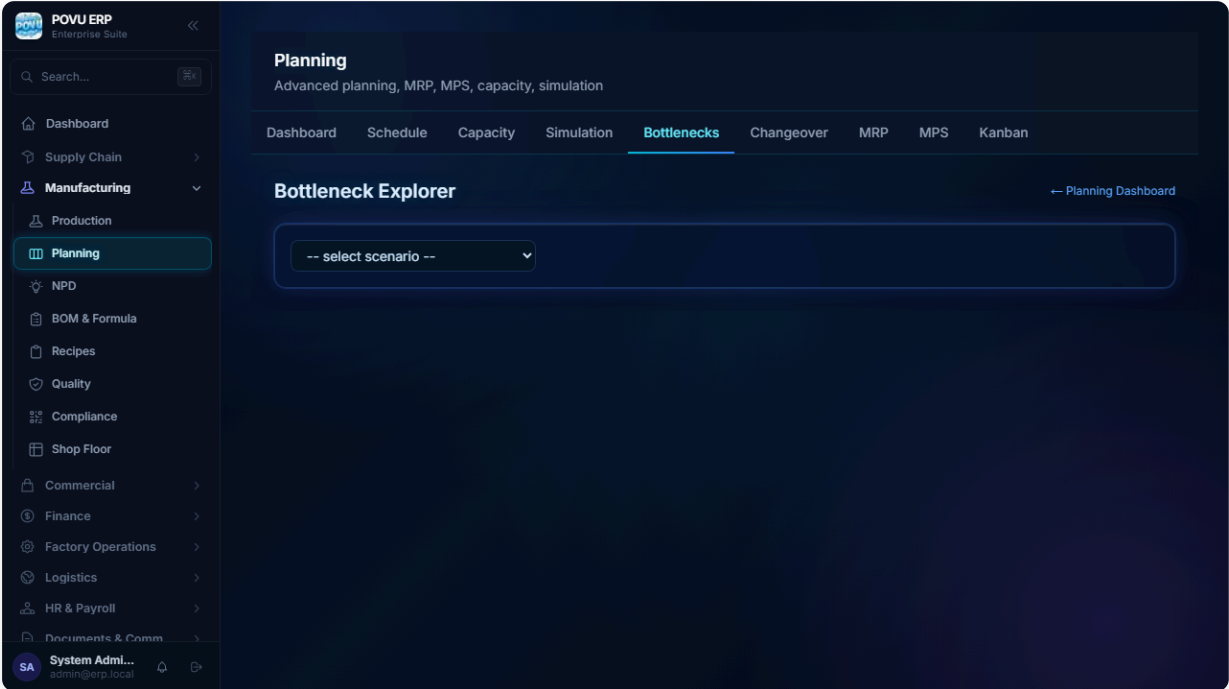
Scenarios are sandbox copies — changes here do not affect production plans until promoted.

Schedule Tab



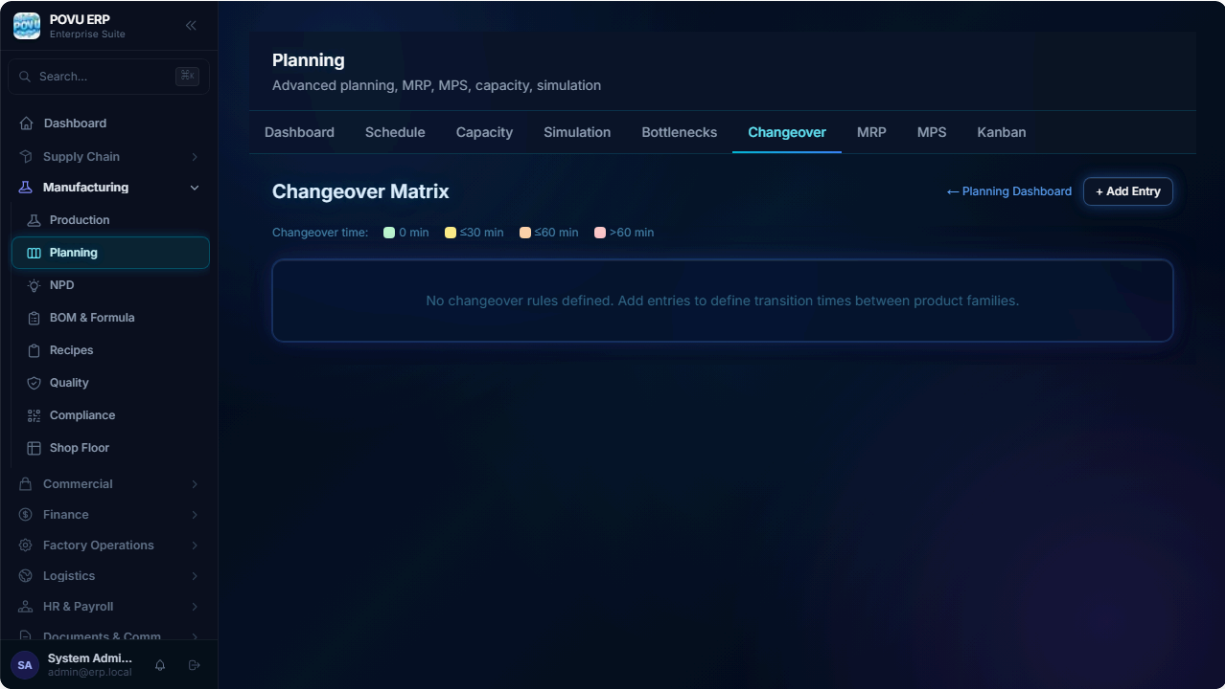
Finite capacity schedule — operations sequenced across work centers.

Bottlenecks Tab



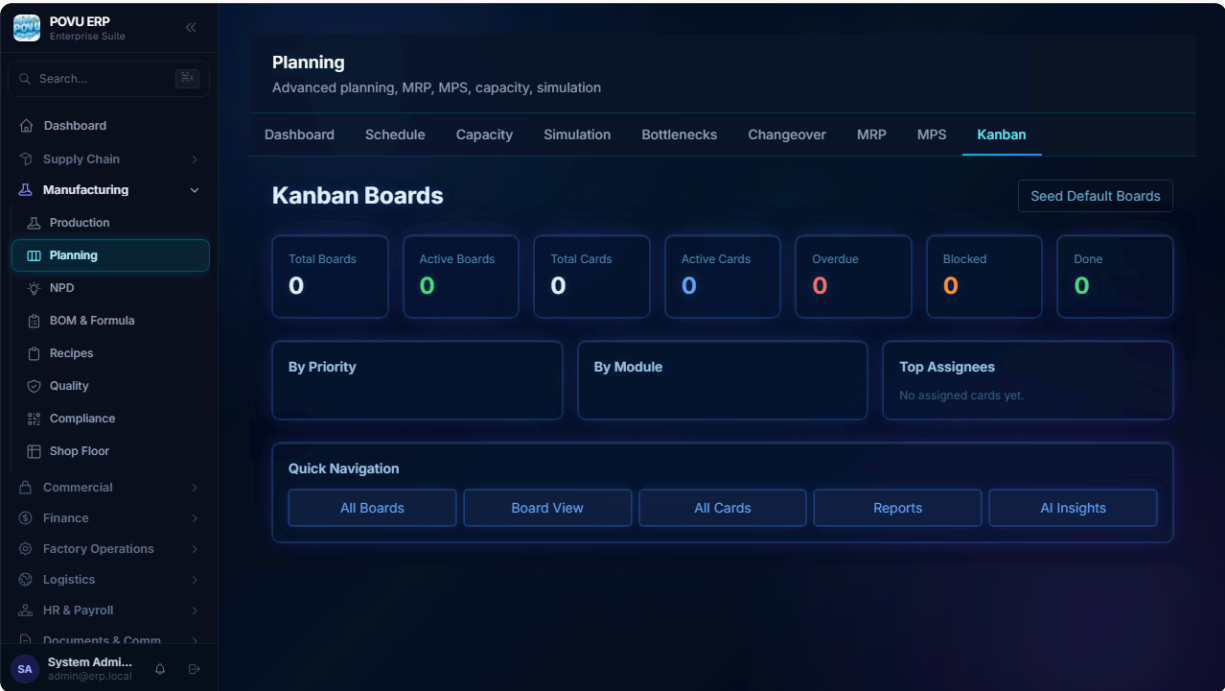
Identifies over-loaded work centers and suggests rebalancing.

Changeover Tab



Changeover time matrix — time to switch between product families on each work center.

Kanban Tab



Kanban board view of production work orders by stage.

7. Shop Floor Operations

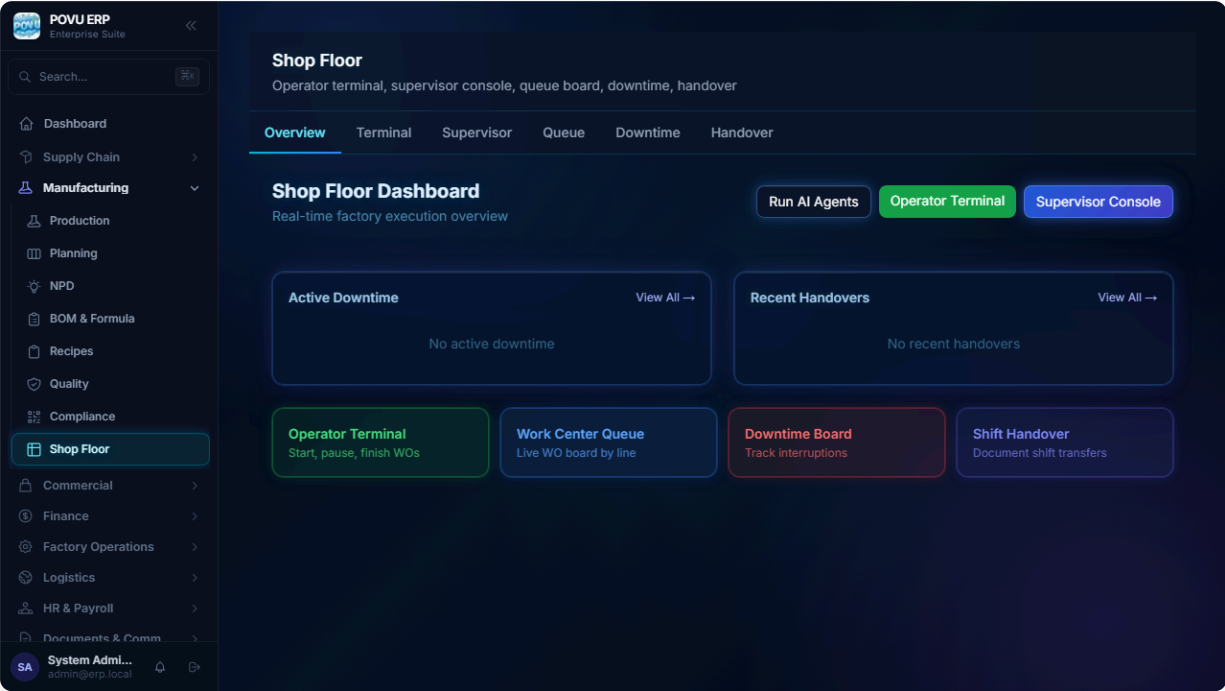
Route: `/dashboard/shop-floor`
Required permission: `production.view`

What it does

Real-time shop floor management — operator terminal, supervisor console, work queue, downtime logging, and shift handover.

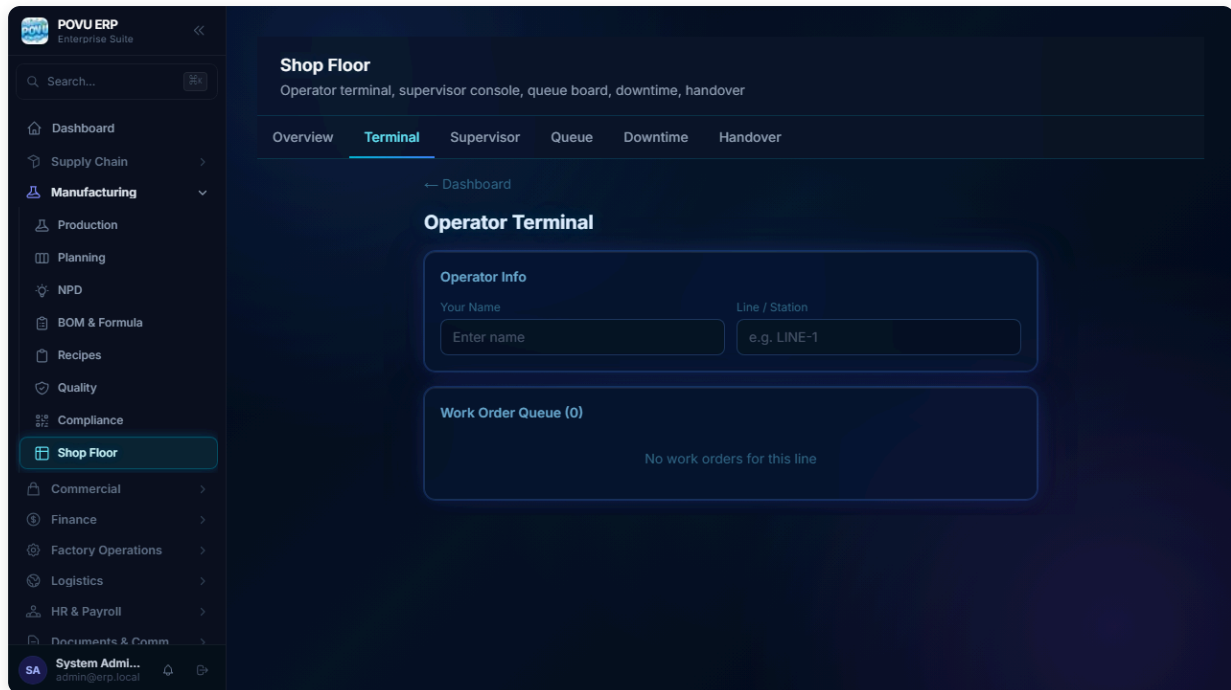
Tabs

Tab	Purpose
Overview	Summary dashboard of all active work orders
Terminal	Operator clock-in/out and work execution
Supervisor	Supervisor override and monitoring console
Queue	Work order queue by work center
Downtime	Shop-floor downtime log
Handover	Shift handover notes



Shop Floor overview — active orders by status and work center.

Terminal Tab

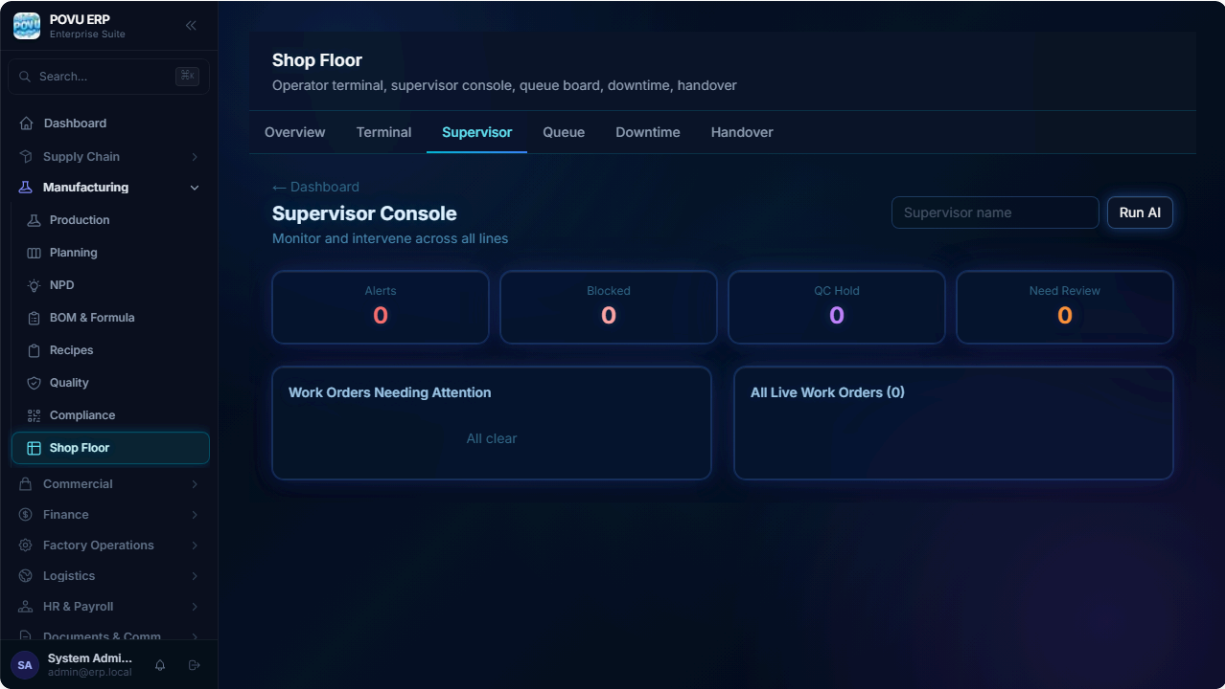


Operator terminal — scan/select work order, start/pause/complete operations.

Operator workflow:

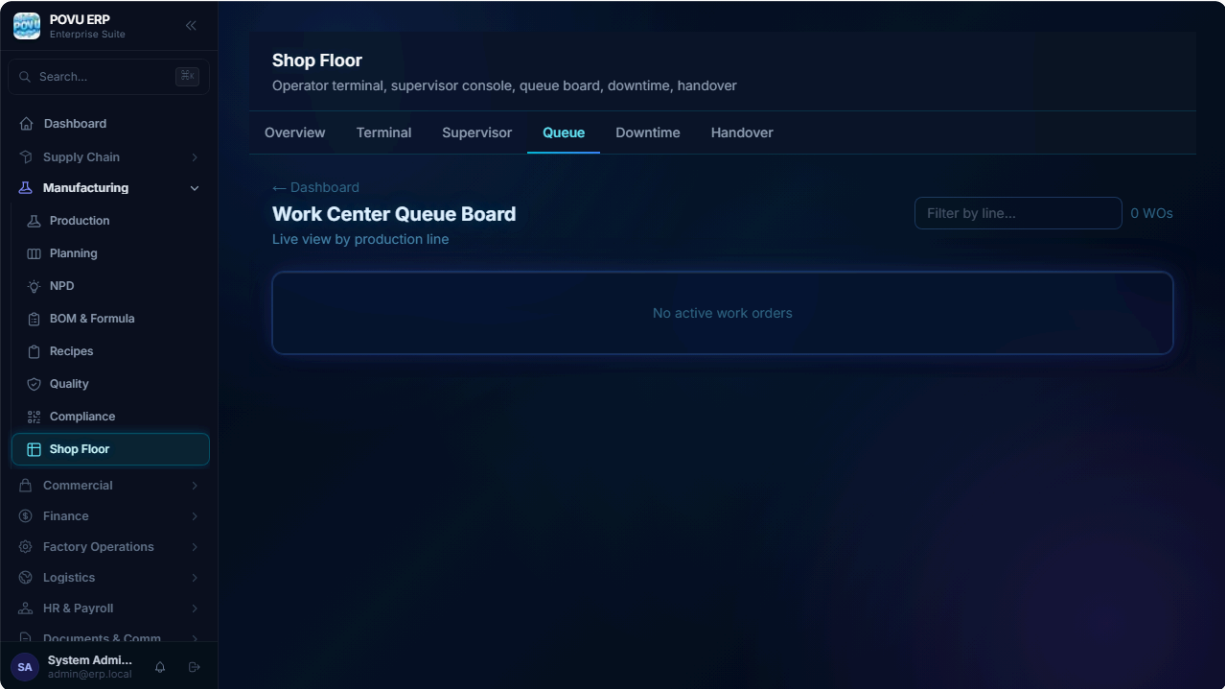
1. Select or scan work order barcode
2. Click **Start** to begin operation
3. Record quantities at checkpoints
4. Click **Pause** to suspend (auto-creates time entry)
5. Click **Complete** when operation finished

Supervisor Tab



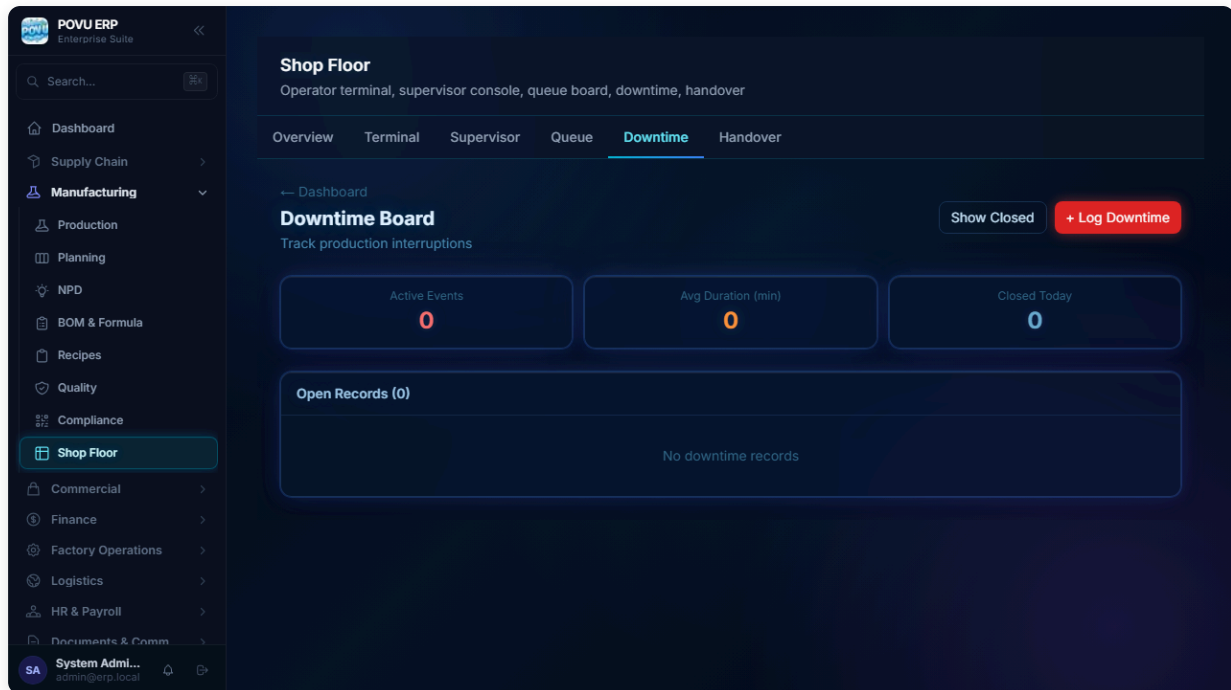
Supervisor console — override decisions, approve exceptions, view operator activity.

Queue Tab



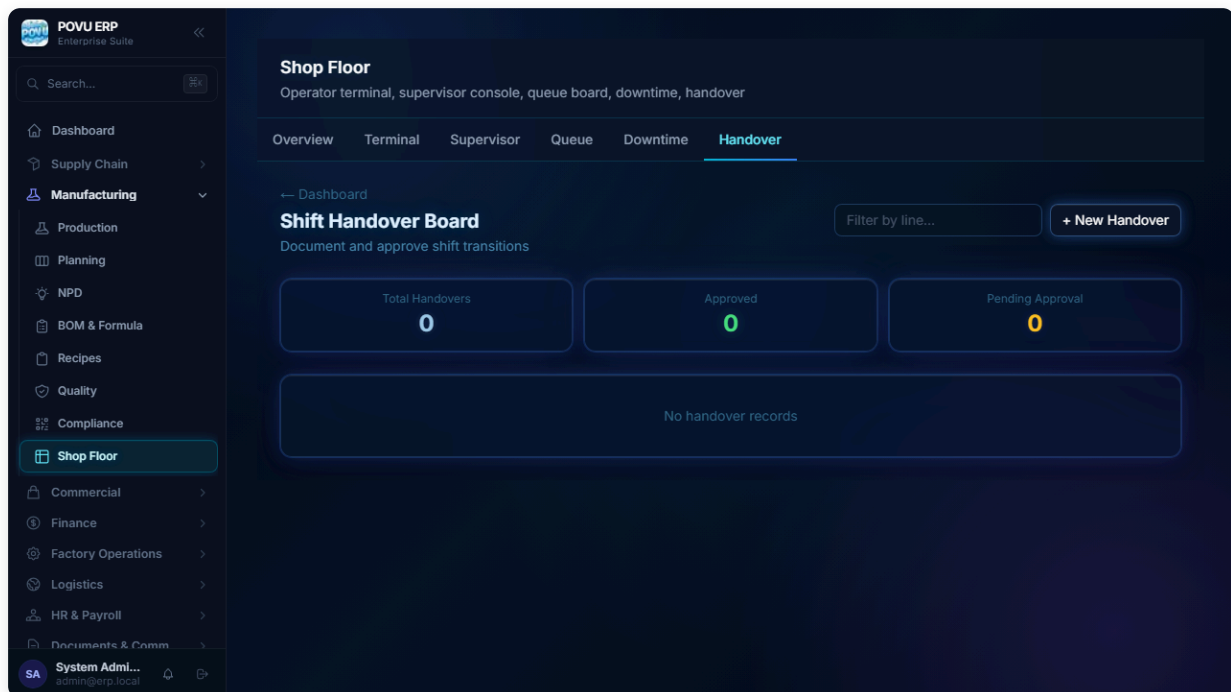
Work order queue per work center — prioritized list of pending operations.

Downtime Tab (Shop Floor)



Log downtime events directly from shop floor with machine and reason code.

Handover Tab



Shift handover — outgoing supervisor records open issues, pending tasks, and notes for incoming shift.

Handover workflow:

1. Open Handover tab at end of shift
2. System auto-populates open work orders and unresolved downtimes
3. Supervisor adds freetext notes

4. Click **Submit Handover** — creates timestamped record
5. Incoming shift supervisor reviews and acknowledges

8. New Product Development (NPD)

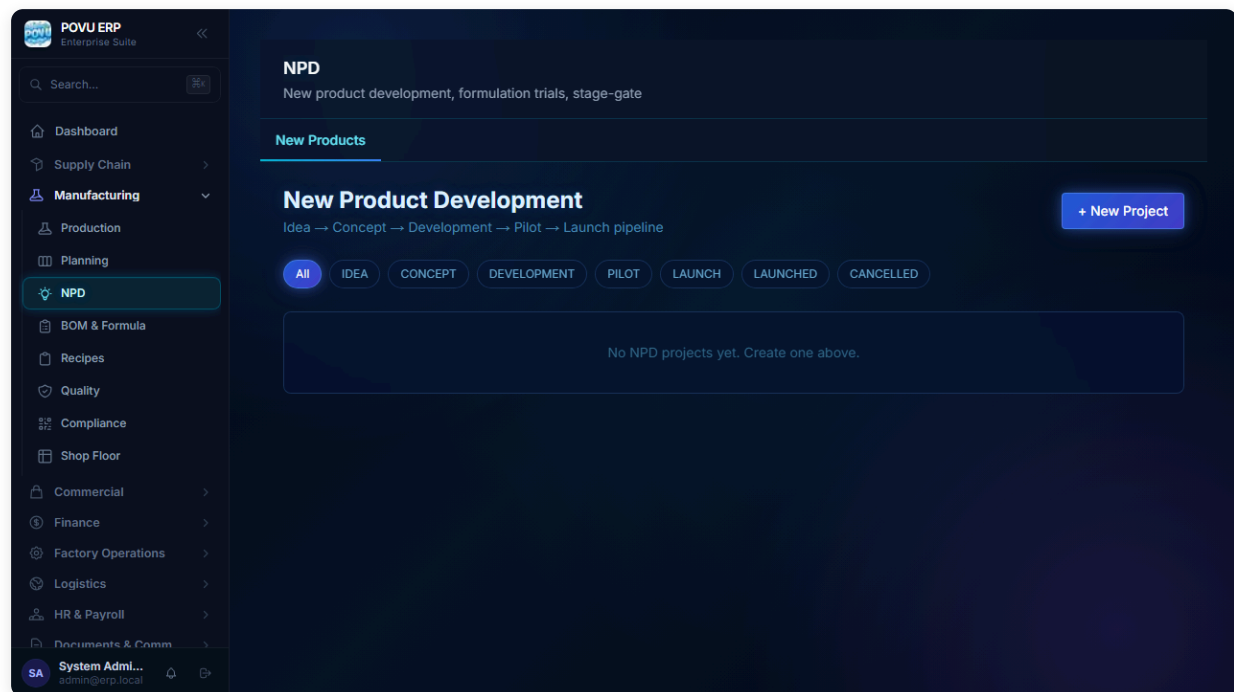
Route: `/dashboard/npd`

Required permission: `npd.view`

What it does

Manages the new product development pipeline using a stage-gate model. From initial concept through trials and commercial launch.

Tabs



NPD project list with stage, owner, and target launch date.

Creating an NPD Project

Click **+ New Project**:

New NPD project form with project name, product category, owner, target dates, and stage gate.

Field	Required	Notes
Project Name	Yes	Descriptive name
Product Category	Yes	Select from categories
Project Owner	Yes	Responsible manager
Target Launch Date	Yes	Commercial target date
Current Stage	Yes	Concept / Development / Trials / Launch
Budget	No	Estimated development budget
Description	No	Free text

Stage Gate Stages

Stage	Meaning
Concept	Initial idea and feasibility
Development	Recipe development and lab trials
Trials	Pilot production and consumer testing
Launch	Commercial production approved
On Hold	Paused
Cancelled	Discontinued

What Happens After Creating a Project

- Project appears in NPD list with CONCEPT stage
- Team members can be assigned
- Linked to Recipe (when trial recipe created) and BOM
- Stage advances when gate criteria met and approved

9. Quality Control

Route: `/dashboard/quality`

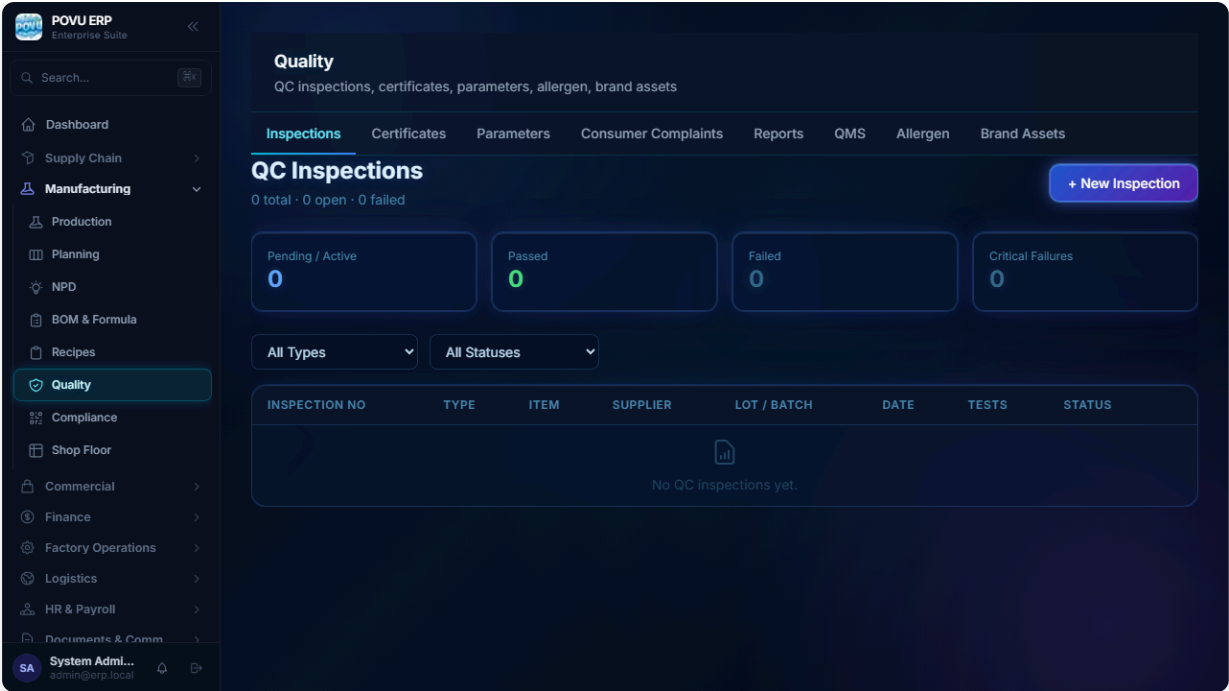
Required permission: `quality.view`

What it does

Manages QC inspections, quality certificates, allergen declarations, consumer complaints, quality parameters, QMS documents, and brand assets.

Tabs

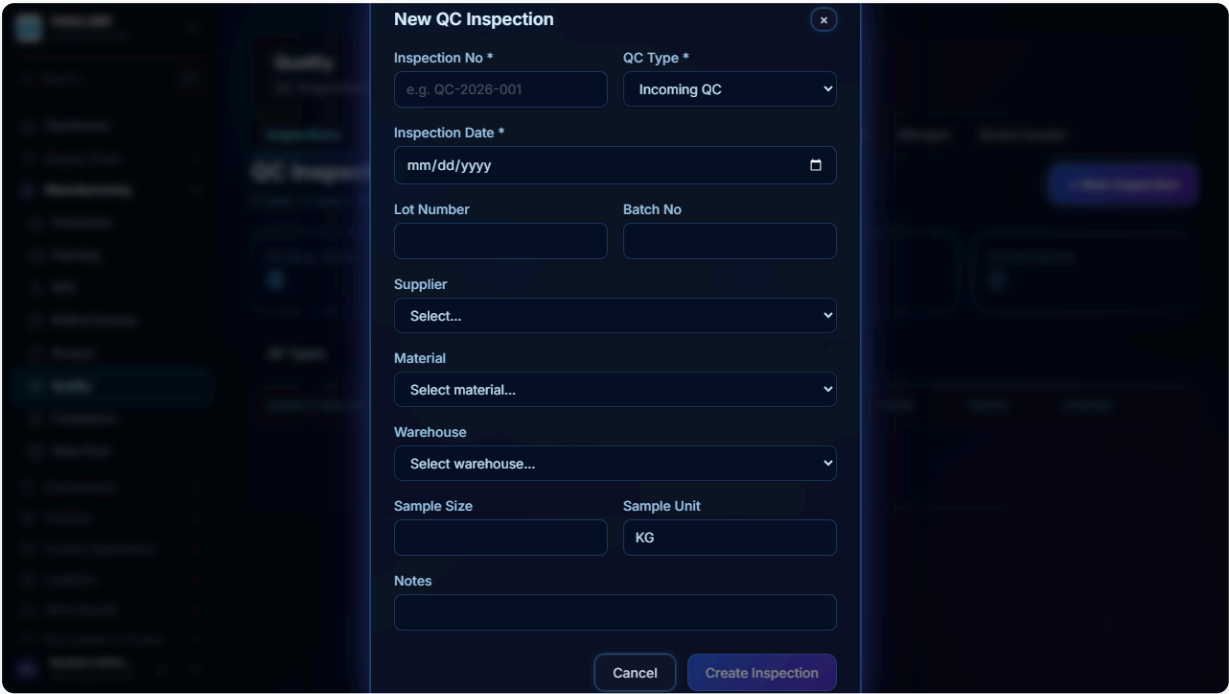
Tab	Purpose
Inspections	QC inspection records
Certificates	Quality certificates per product/batch
Parameters	Quality test parameters configuration
Consumer Complaints	Customer complaint log and tracking
Reports	QC summary and trend reports
QMS	Quality management system documents
Allergen	Allergen declaration matrix
Brand Assets	Brand standards and packaging



Inspections list with type, product, batch, result, and status columns.

Creating an Inspection

Click + New Inspection:



Field	Required	Notes
Inspection Type	Yes	Incoming / In-Process / Final / Outgoing
Product	Yes	Product being inspected
Batch/Lot	No	Link to batch if applicable
Work Order	No	Link to production work order
Inspector	Yes	User performing inspection
Scheduled Date	Yes	Planned inspection date

After creation, add test results against the configured parameters.

Inspection Results — Decision

Decision	Meaning
PASS	Product meets all specifications
FAIL	Product outside specification — triggers NCMR
CONDITIONAL	Passes with deviations noted
PENDING	Results not yet complete

Certificates Tab

POVU ERP
Enterprise Suite

Search...

Dashboard
Supply Chain
Manufacturing
Production
Planning
NPD
BOM & Formula
Recipes
Quality
Compliance
Shop Floor
Commercial
Finance
Factory Operations
Logistics
HR & Payroll
Documents & Comm
System Admin...

Quality
QC inspections, certificates, parameters, allergen, brand assets

Inspections **Certificates** Parameters Consumer Complaints Reports QMS Allergen Brand Assets

Regulatory Certificate Register
KEBS, HALAL, ISO, FSCC, NAFDAC and all regulatory certifications

+ Add Certificate

SyntaxError: Unexpected token '<', '<DOCTYPE '...' is not valid JSON

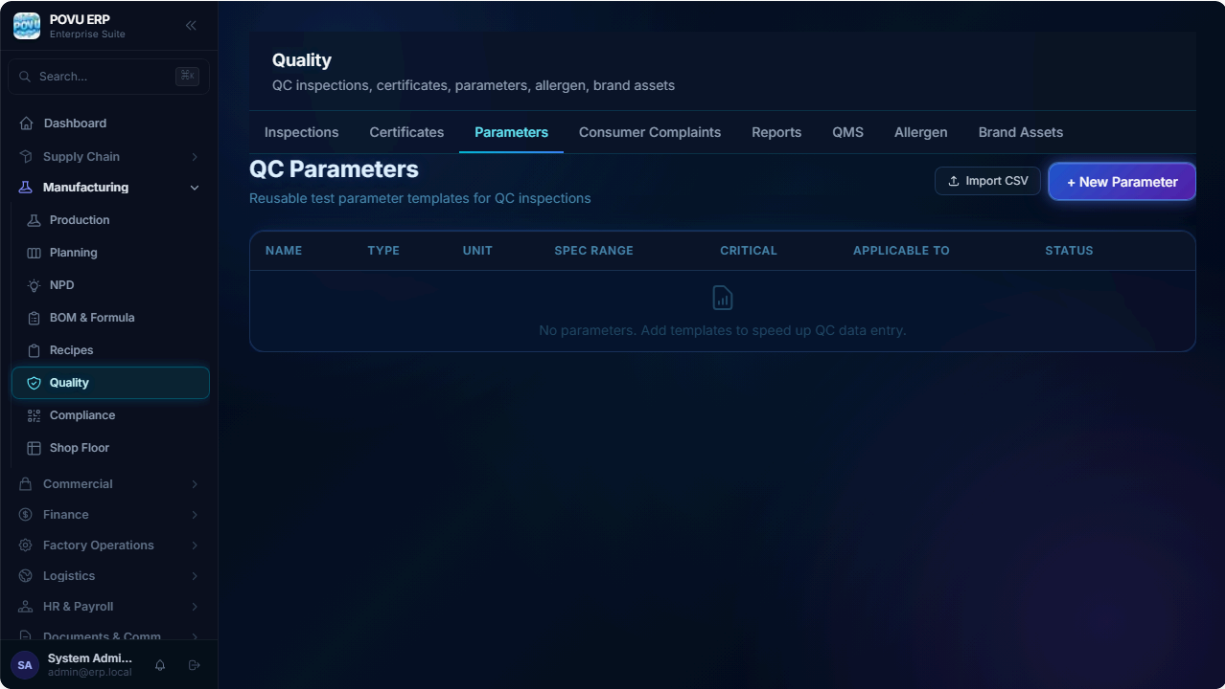
Status: All Entity Type: All ☐ Expiring / Expired only [Clear](#)

Certificates (0)

No certificates found.

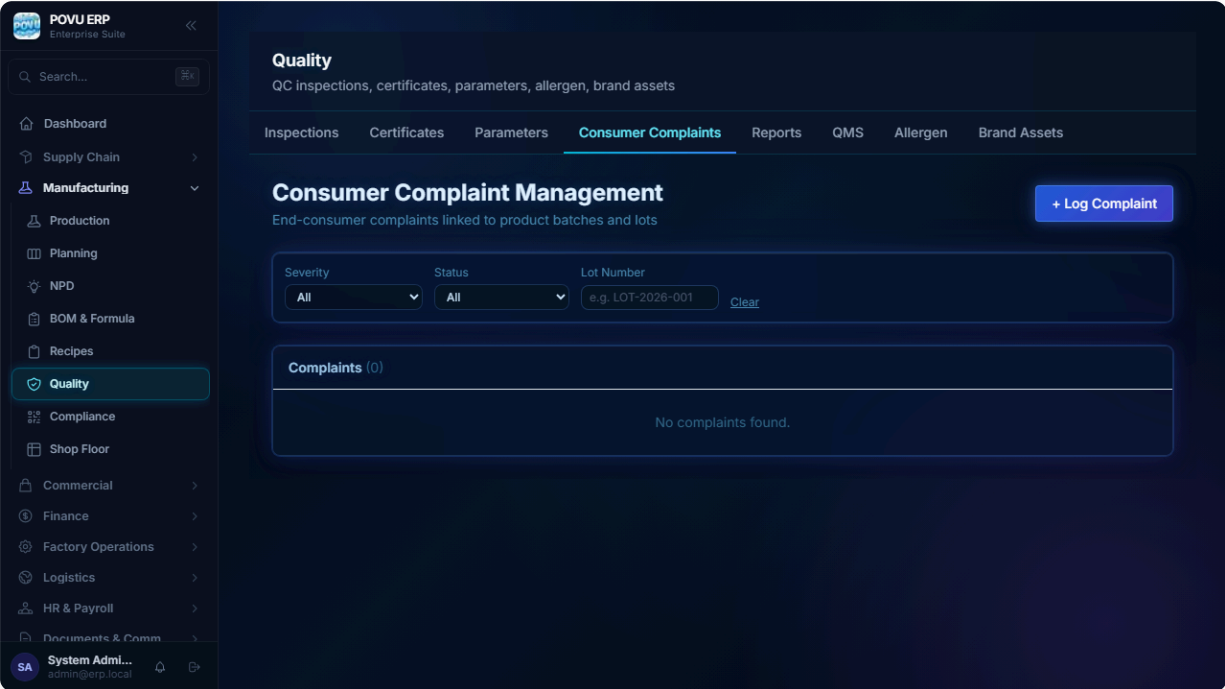
Quality certificates — COA (Certificates of Analysis) per product and batch.

Parameters Tab



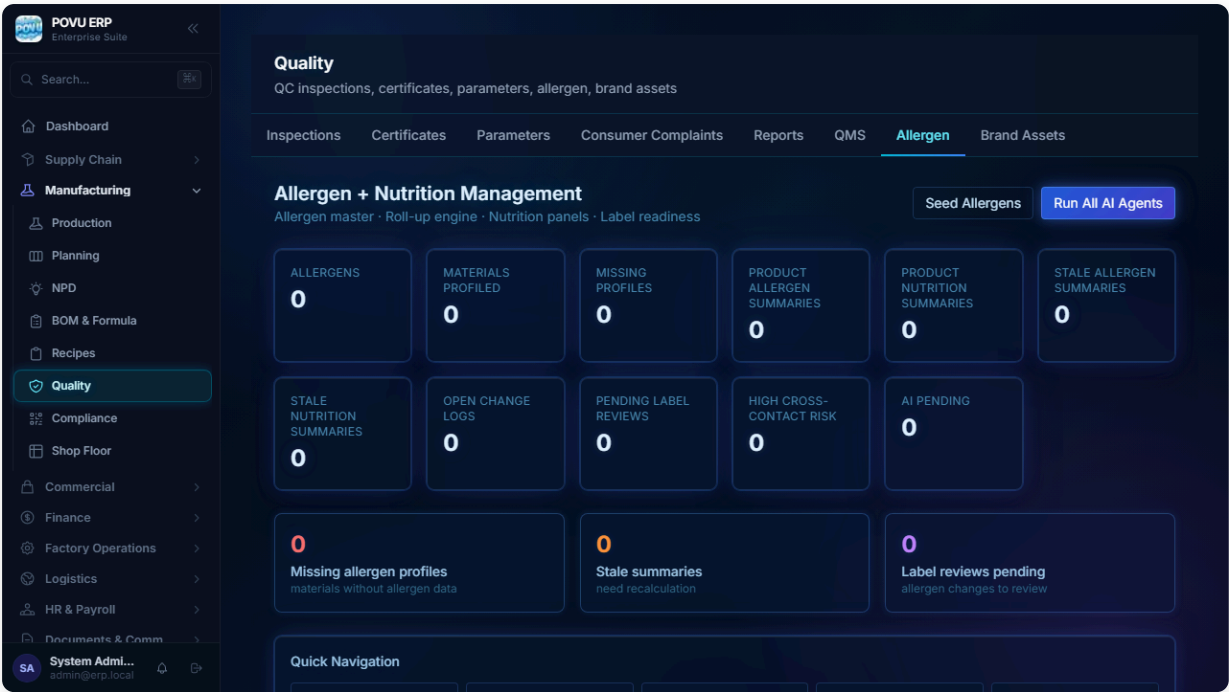
Quality test parameters — define specs (min/max/target) for each product-test combination.

Consumer Complaints Tab



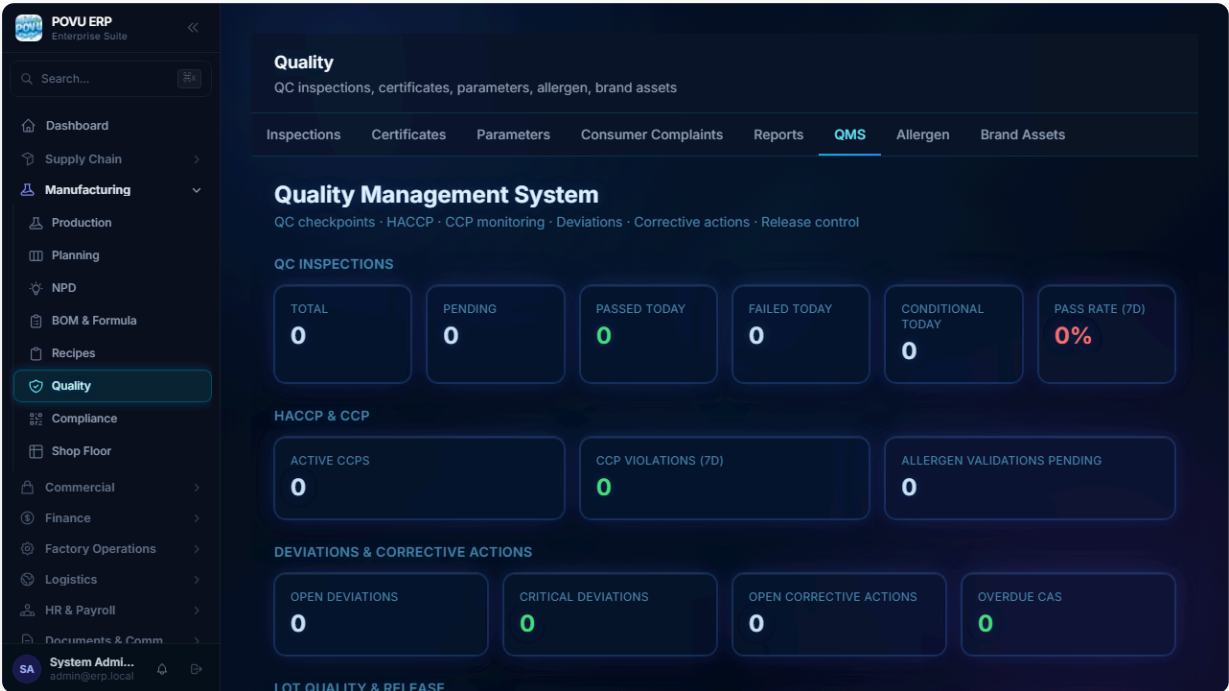
Consumer complaint register — log, categorize, investigate, and close complaints.

Allergen Tab



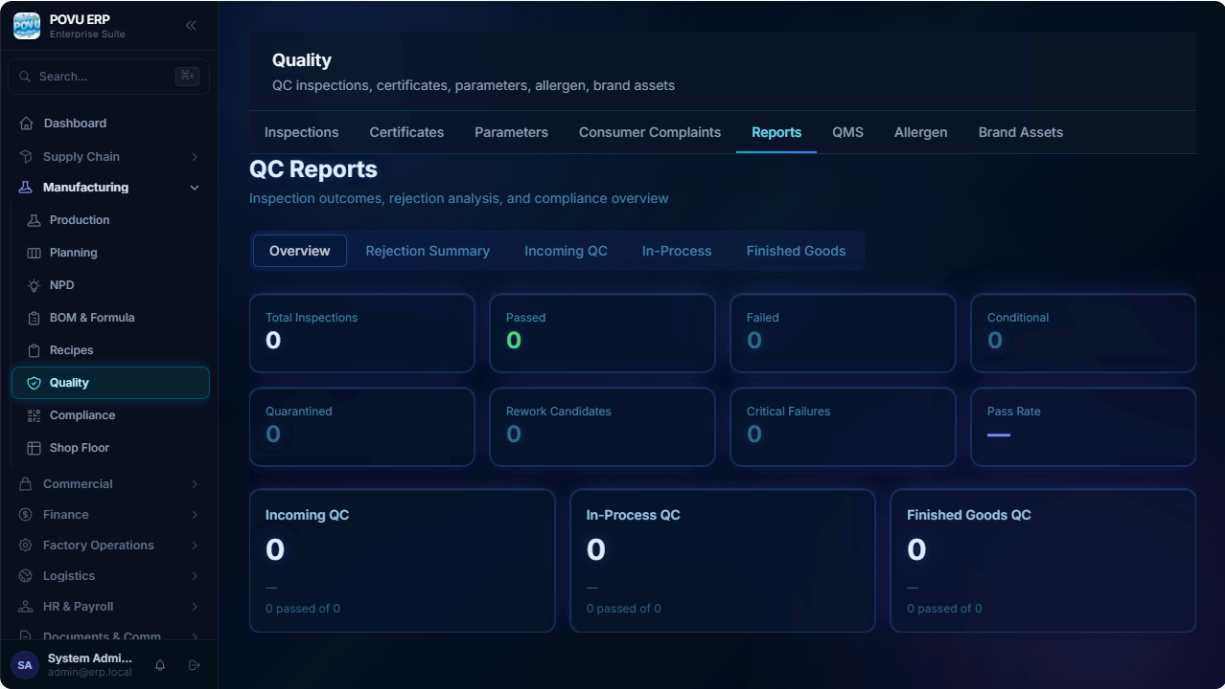
Allergen matrix — cross-reference products with allergen presence and contamination risk.

QMS Tab



QMS document library — SOPs, work instructions, and quality procedures.

Reports Tab



QC trend charts and summary statistics by product, line, and period.

10. Compliance & Labelling

Route: `/dashboard/compliance`
Required permission: `gs1.view`

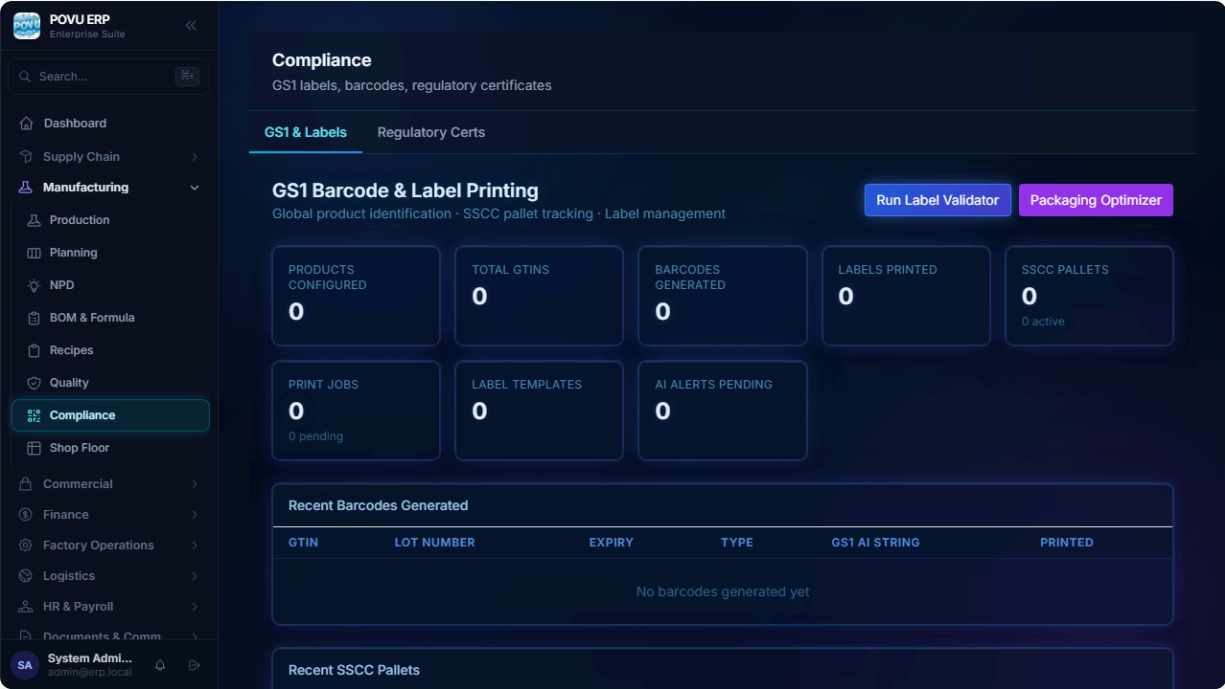
What it does

Manages GS1 barcode label generation and regulatory certificate tracking for export and local market compliance.

Tabs

Tab	Purpose
GS1 Labels	Generate and print GS1-compliant barcodes
Regulatory Certs	Track regulatory permits and certificates

GS1 Labels Tab



GS1 label generation — select product, batch, and label template to generate barcodes.

Regulatory Certs Tab



Regulatory certificates register — status, expiry tracking, and renewal alerts.

Status filter:



Filter by: Active / Expired / Expiring Soon / Pending

Entity type filter:



Filter by entity: Product / Facility / Ingredient / Process

Adding a Certificate

Compliance
GS1 labels, barcodes, regulatory certificates

GS1 & Labels **Regulatory Certs**

Regulatory Certificate Register
KEBS, HALAL, ISO, FSCC, NAFDAC and all regulatory certifications

[+ Add Certificate](#)

SyntaxError: Unexpected token '<', '<!DOCTYPE "... is not valid JSON

Status: All Entity Type: All ☐ Expiring / Expired only [Clear](#)

Register Certificate

Entity Name * Authority *

Entity Type Certificate Number Issued Date

Expiry Date Country Status

Document URL

Field	Required	Notes
Certificate Name	Yes	e.g. "KEBS Approval — Tomato Sauce"
Issuing Authority	Yes	e.g. KEBS, KEPHIS, HISA
Entity Type	Yes	Product / Facility / Ingredient
Reference No	Yes	Official certificate/permit number
Issue Date	Yes	Date issued
Expiry Date	Yes	Renewal required before this date
Status	Yes	Active / Expired / Pending
Product / Facility	No	Link to relevant entity
Document	No	Upload PDF/scan of certificate

Expiry Alerts

System flags certificates expiring within 90 days in the KPI strip at top of Regulatory Certs tab.

11. Common Mistakes & Troubleshooting

Problem	Cause	Fix
Can't edit recipe	Status is APPROVED	Create new version as DRAFT; cannot reverse APPROVED
Work order won't confirm	No approved BOM linked	Approve BOM first; then link to work order
MRP recommends nothing	Demand horizon too short or no open sales orders	Extend horizon; check sales order backlog
OEE shows 0%	No work orders in COMPLETED status	Complete work orders; ensure times entered
QC inspection missing parameters	No parameters configured for this product	Add parameters in Quality → Parameters tab
Shop Floor terminal shows empty queue	Work orders not RELEASED	Release work orders from Production → Work Orders tab
Regulatory cert not showing	Certificate expired filter active	Clear status filter or update status to Active

12. Related Modules

This Action	Connects To
Recipe → production use	Production → Work Orders (BOM + Recipe linked)
Work Order completion	Inventory → Stock (FG increase + RM consumption)
MRP recommendations	Procurement → Purchase Requisitions
QC Fail → NCMR	Quality → Consumer Complaints or Inventory → Quarantine
NPD to commercial	Recipes → BOM → Production Plan
Allergen declarations	Compliance → Regulatory Certs (certificate links to allergen certs)
Batch tracking	Inventory → Traceability and Serials modules

End of Manufacturing Manual v2